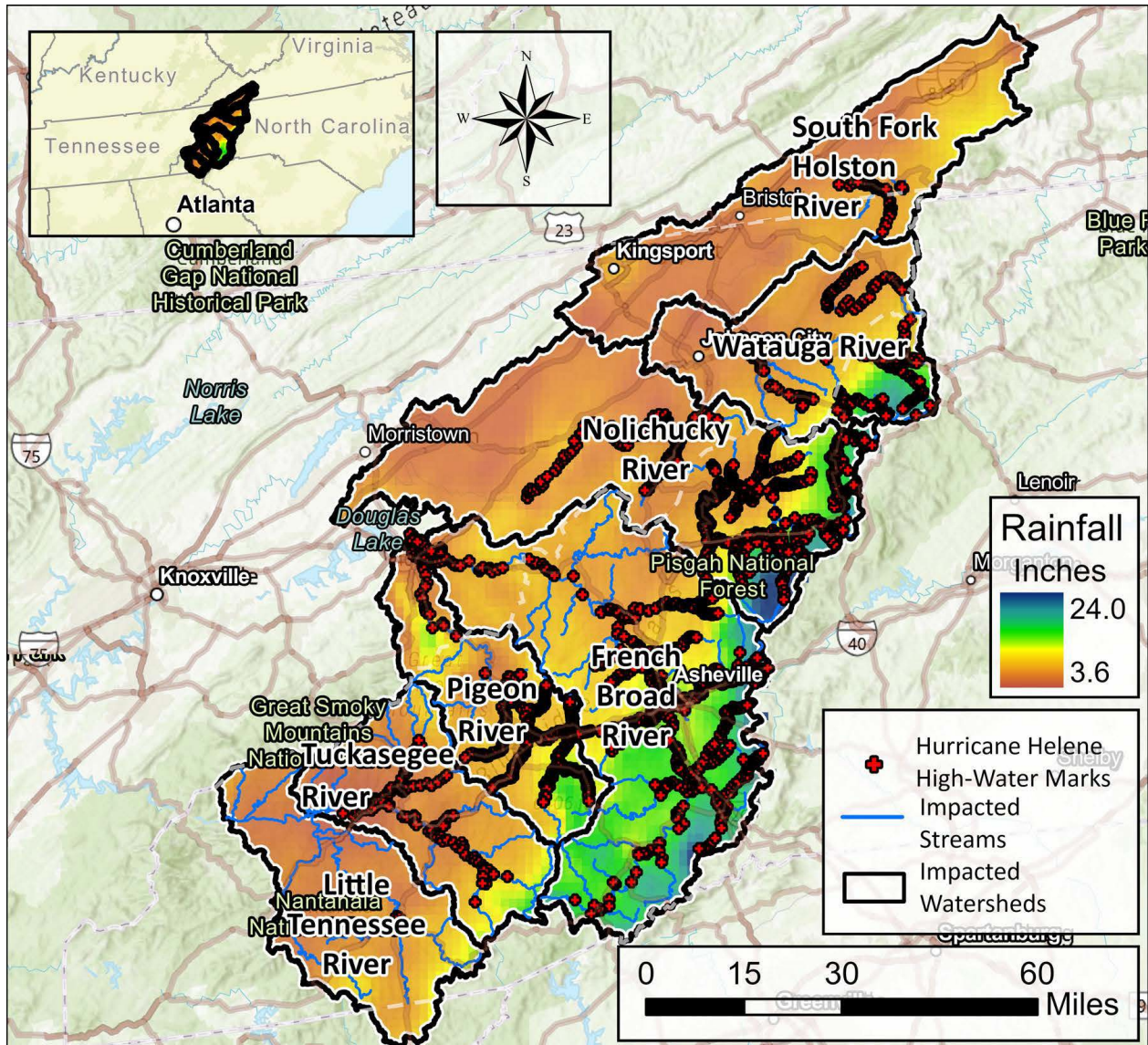




U.S. ARMY CORPS OF ENGINEERS
NASHVILLE DISTRICT

Post Flood Report

Hurricane Helene Flood – September 2024



Prepared by
US Army Corps of Engineers
Nashville District
Water Resources Section

Summary

This Nashville District U.S. Army Corps of Engineers (USACE) technical report is intended to provide local communities and agencies with a thorough understanding of the Hurricane Helene Flood Event in September 2024. This Post-Flood Report documents the meteorologic conditions of the storm, the hydraulic characteristics of the flood event, the factors that contributed to the severity of the flood, and information on the collected high-water marks after the flood. This report will serve the Nashville District and the region as a reference tool for future project planning and will advance the Corps' mission for the protection of life and property in the Cumberland River and Tennessee River Basins. This report is also intended to improve public awareness of this event, thereby potentially reducing the consequences of future flood events.

Hurricane Helene made landfall in Florida as a Category Four hurricane on the evening of September 26. Preceding the hurricane, a slow-moving front produced heavy rainfall across the Tennessee Valley and the southern Appalachian Mountains on September 25 - 26. This storm front produced 2 to 7 inches of rain in East Tennessee and western North Carolina before rainfall associated with Helene hit the area on September 27. By midnight on September 27, the 3-day total rainfall accumulation was between 10 to over 24 inches in western North Carolina, causing extreme flooding along many rivers and watersheds. Many of these rivers including the French Broad River, Pigeon River, and Nolichucky River feed into East Tennessee from North Carolina, so although less rainfall occurred there, the flooding was still extreme.

Significant, widespread, and record-breaking flooding occurred across the mountains of western North Carolina, eastern Tennessee, and southern Virginia, bringing catastrophic changes to the landscape in some areas. The disaster damaged or destroyed numerous roads and bridges, dislodged buildings, and left communities devastated. The images on the next page give just a couple of the areas impacted by the flooding. The region's steep, mountainous terrain amplified the crisis as flash flooding struck with little warning, allowing minimal time for evacuation. Normally small, shallow streams transformed into vast, valley-spanning floodplains, highlighting the unprecedented scale of the event.

The USACE coordinated with local, state, and federal representatives and agencies to complete a post-flood mission in the Nashville District area of responsibility (AOR). The goal of this mission was to collect High-Water Marks (HWMs) and gather as much flood-related data as possible in the impacted areas. 2,587 HWMs were collected and surveyed as part of this mission. The HWM collection teams covered 19 counties and 115 streams across Tennessee, North Carolina, and Virginia.

HURRICANE HELENE – POST FLOOD REPORT



Table of Contents

Summary.....	ii
1 Introduction	1
1.1 Authority and Appropriations	1
1.2 Scope.....	1
1.3 Description and History of Area.....	2
1.4 Flood Event	2
2 Coordination.....	4
2.1 Interagency Coordination	4
2.2 USACE Coordination	5
3 Hurricane Helene Event.....	6
3.1 Historic Storms and Floods.....	6
3.2 Antecedent Conditions	9
3.3 Hurricane Helene Rainfall.....	13
3.4 Hurricane Helene Peak Stage and Flow Data	21
3.4.1 Pigeon River Watershed.....	23
3.4.2 French Broad River Watershed.....	26
3.4.3 Nolichucky River Watershed	30
3.4.4 Watauga, South Fork Holston, Tuckasegee, and Little Tennessee River Watersheds	33
4 Field Data Collection.....	38
4.1 High-Water Mark Data.....	38
4.1.1 High-Water Mark Significance.....	40
4.1.2 High-Water Mark Pre-Deployment Preparation.....	42
4.1.3 High-Water Mark Data Collection Process	44
4.2 Surveying.....	46
4.2.1 Survey Equipment and Accuracy.....	46
4.2.2 Survey and HWM Coordinate Systems.....	47
4.3 First-Hand Accounts of Hurricane Helene	47
4.4 Post-Processing and Final Database	51
5 Conclusion.....	51
6 References	52
Appendix A: Hurricane Helene Flood Photographs.....	53
Appendix B: High-Water Mark Data Collection Table - Example.....	58

List of Figures

Figure 1.1: Report County Map.....	3
Figure 3.1: NWS Gages Map	8
Figure 3.2: Antecedent Storm – September 24, 1200, EST – September 26, 1200, EST – Rainfall Totals...10	
Figure 3.3: USGS Gage 03456991 Pigeon River near Canton, NC – Stage Hydrograph – Antecedent Stage	11
Figure 3.4: USGS Gage 03451500 French Broad River at Asheville, NC – Stage Hydrograph – Antecedent Stage	12
Figure 3.5: USGS Gage 03465500 Nolichucky River at Embreeville, TN – Stage Hydrograph – Antecedent Stage	12
Figure 3.6: USGS Gage 03513000 Tuckasegee River at Bryson City, NC – Stage Hydrograph – Antecedent Stage	13
Figure 3.7: Hurricane Helene Rainfall – September 24, 1200, EST – September 27, 1200, EST – 3-Day Rainfall Totals.....	14
Figure 3.8: NOAA Hurricane Helene 3-day Rainfall Annual Exceedance Probabilities (National Oceanic and Atmospheric Administration, 2024).....	15
Figure 3.9: CoCoRaHS 3-day Hurricane Helene Rainfall Totals (Community Collaborative Rain, Hail, and Snow Network, 2024)	16
Figure 3.10: USGS Gage 03451500 French Broad River at Asheville, NC - 3-day Precipitation Total.....	17
Figure 3.11: USGS Gage 03451000 Swannanoa River at Biltmore, NC - 3-day Precipitation Total	17
Figure 3.12: USGS Rain Gage French Broad River near Fletcher, NC - 3-day precipitation total	18
Figure 3.13: USGS Rain Gage Slicking Mtn near Rocky Bottom, SC - 3-day precipitation total.....	18
Figure 3.14: USGS Gage 03456500 East Fork Pigeon River near Canton, NC - 3-day precipitation total ...	19
Figure 3.15: USGS Gage 03479000 Watauga River near Sugar Grove, NC - 3-day precipitation total.....	19
Figure 3.16: Hurricane Helene - Peak Stage Gage Map.....	22
Figure 3.17: USGS Gage 03461500 Pigeon River at Newport TN – Stage Hydrograph.....	23
Figure 3.18: Pigeon River Watershed - Peak Stages	25
Figure 3.19: USGS Gage 03447687 French Broad River near Fletcher, NC – Stage Hydrograph.....	26
Figure 3.20: USGS Gage 03451500 French Broad River at Asheville, NC – Stage Hydrograph.....	27
Figure 3.21: USGS Gage 03455000 French Broad River near Newport, TN – Stage Hydrograph	27
Figure 3.22: French Broad River - Peak Stages	29
Figure 3.23: USGS Gage 03467609 Nolichucky River near Lowland, TN – Stage Graph.....	30
Figure 3.24: Nolichucky River - Peak Stages.....	32
Figure 3.25: USGS Gage 03486000 Watauga River at Elizabethton, TN – Stage Hydrograph	33
Figure 3.26: USGS Gage 03473000 South Fork Holston River near Damascus, VA – Stage Graph.....	34
Figure 3.27: USGS Gage 03510577 Tuckasegee River at Barker’s Creek, NC – Stage Graph.....	35
Figure 3.28: USGS Gage 03513000 Tuckasegee River at Bryson City, NC – Stage Graph.....	35
Figure 3.29: Watauga, South Fork Holston, and Tuckasegee River - Peak Stage.....	37
Figure 4.1: Hurricane Helene – High-Water Mark (HWM) Overview Map.....	39
Figure 4.2: Mud Line on Building along the Swannanoa River, NC.....	40
Figure 4.3: Debris Snag in Fence along the Doe River, TN	41
Figure 4.4: Mud Line on Building along the Pigeon River, TN.....	41

HURRICANE HELENE – POST FLOOD REPORT

Figure 4.5: Mud Line on Building along the French Broad River, NC	42
Figure 4.6: NOAA Aerial Imagery Before and After - Asheville, NC (National Oceanic and Atmospheric Administration, 2024).....	43
Figure 4.7: High-Water Mark Data Collection Sheet - Front.....	44
Figure 4.8: High-Water Mark Data Collection Sheet - Back	45
Figure 4.9: Trimble R12i Tilt of Survey Unit	47
Figure 4.10: Flooding in Asheville, NC	48
Figure 4.11: Flooding in Candler, NC	49
Figure 4.12: Dillingham Rd - Barnardsville, NC.....	50
Figure 4.13: Johnson County Airport - Johnson County, TN.....	50
Figure A.1: Flood Damage on bridge in Asheville, NC.....	53
Figure A.2: Flood Damage on billboard Asheville, NC.....	53
Figure A.3: Flood Damage in Swannanoa River near Swannanoa, NC.....	54
Figure A.4: Pigeon River floodwaters discharging from Watersville Dam.....	54
Figure A.5: Damage done to US HWY 421 near Mountain City, TN.....	55
Figure A.6: Damage done to Rhodes Road in Edneyville, NC	55
Figure A.7: Damage done to Rhodes Road in Edneyville, NC	56
Figure A.8: Flood damage at Peppertree Resort in Maggie Valley, NC.....	56
Figure A.9: Flood damage in recreational park.....	57
Figure A.10: Lake Craig Dam Failure on Swannanoa River near Asheville, NC.....	57

List of Tables

Table 3.1: NWS Gage Table – Major Floods	7
Table 3.2: Hurricane Helene – Precipitation Totals	20
Table 3.3: Pigeon River Watershed - Gage Data	24
Table 3.4: French Broad River Watershed - Gage Data	28
Table 3.5: Nolichucky River Watershed - Gage Data.....	31
Table 3.6: Watauga, South Fork Holston, and Tuckasegee River Watersheds - Gage Data	36
Table 4.1: Number of HWMs Collected per Watershed.....	38
Table 4.2: High-water Mark Quality Chart.....	45
Table B.1: HWM Field Data Collection Log - Example.....	58
Table B.2: HWM Post-Processing Data - Example	59

1 Introduction

1.1 Authority and Appropriations

This report was prepared under the authority of Class 210 of Public Law (P.L.) 84-99, Emergency Response to Natural Disasters, in accordance with Engineer Regulation (ER) 500-1-1. Class 210 refers to a specific funding category within Public Law 84-99, which authorizes the U.S. Army Corps of Engineers to provide emergency assistance during flood events, specifically allocating funds for direct flood fighting operations and technical assistance to support state and local efforts during a flood crisis. For Hurricane Helene, Class 210 funds were used for data collection to record flood information unique to the basin that would be irretrievably lost if not collected immediately after the flood. Data collection was mostly limited to record or near-record flood levels and was supplemental to state and local efforts.

In accordance with ER 500-1-1, paragraph 4-3:

- (1) Acceptable data collection activities include: ground and aerial photography (not to include controlled aerial photography or map preparation); setting high-water marks for future recovery and documentation under other authorities; and basic hydrologic data not obtainable under other programs, from USACE project activities or from other agencies.
- (2) Detailed flood damage surveys or analysis, collection of economic data, hydrologic studies, compilation of comprehensive flood data, sedimentation surveys, collection of data on socio-economic impacts, and detailed frequency analysis cannot be undertaken using this authority.
- (3) Any data collected should be maintained at the district as a permanent reference file. All data collection activities should be substantially complete when the emergency response/Post Flood Response activities are complete.

Pursuant to this funding class, high-water mark data was collected, post-processed, and documented for impacted basins in Tennessee, Virginia, and North Carolina within the Nashville District area of responsibility. Although this mission was not a Federal Emergency Management Agency (FEMA) Mission Assignment, the requirements for utilizing emergency management (EM) funds necessitate daily situational reports (SITREPs) detailing the personnel active/deployed, completed tasks, and upcoming activities.

1.2 Scope

The purpose of the Hurricane Helene Post Flood Technical Report is to provide local communities and agencies with a ready reference for information related to the record flood event in September 2024 which affected East Tennessee, southwest Virginia, and western North Carolina. This report will serve the Nashville District and the regions as a reference tool for future project planning and will advance the Corps' mission for the protection of life and property in the affected areas. This report investigates and documents the meteorologic conditions of the storm, the hydraulic characteristics of the event, and the factors which contributed to the severity of the event. An evaluation of the severity of the flood was conducted including the surveying of high-water marks (HWMs) across the affected regions.

HURRICANE HELENE – POST FLOOD REPORT

1.3 Description and History of Area

Situated amidst the Blue Ridge Mountains are many counties throughout western North Carolina and eastern Tennessee. The counties where the Nashville District collected HWMs from the Hurricane Helene flood are displayed in Figure 1.1 below. The hydrology in these regions is indicative of the steep mountain topography with high peaks, deep valleys, and many rivers which leaves these areas prone to flooding. Much of the development and roadways in these areas are in the narrow floodplains and along the streams in the region. Therefore, the disaster damaged or destroyed numerous roads and bridges, dislodged buildings, and left communities devastated. The region's steep, mountainous terrain amplified the crisis as flash flooding struck with little warning, allowing minimal time for evacuation. Normally small, shallow streams transformed into vast, valley-spanning floodplains.

1.4 Flood Event

This Post Flood Report is an evaluation of the flood event impacts from Hurricane Helene on major streams and tributaries in East Tennessee, southwest Virginia, and western North Carolina, including the Pigeon River, Nolichucky River, Watauga River, and French Broad River basins.

Hurricane Helene was a Category Four storm when it made landfall in the Florida Gulf Coast on September 26, 2024. However, the storm's largest impacts were felt inland across the Southern Appalachians where widespread and catastrophic flooding occurred. The hurricane moved over Appalachia on September 27 and dissipated to the northwest by September 29, resulting in the declaration of state of emergencies for North Carolina, Tennessee, and Virginia after the historic storm.

The storm left entire communities in complete ruin, with damaged roadways, destroyed bridges, and impaired critical public infrastructure. Major rivers such as the Pigeon, French Broad, Nolichucky, and Swannanoa, saw record-breaking stage and flow records. Some streams were re-routed entirely due to the sheer amount of volume and force of floodwaters. According to State emergency management agencies, there were 99 confirmed fatalities in the Nashville District AOR. Hurricane Helene left behind widespread devastation, with severe flooding, extensive property damage, and disrupted communities, highlighting the immense impact of powerful storms on vulnerable regions.

HURRICANE HELENE – POST FLOOD REPORT

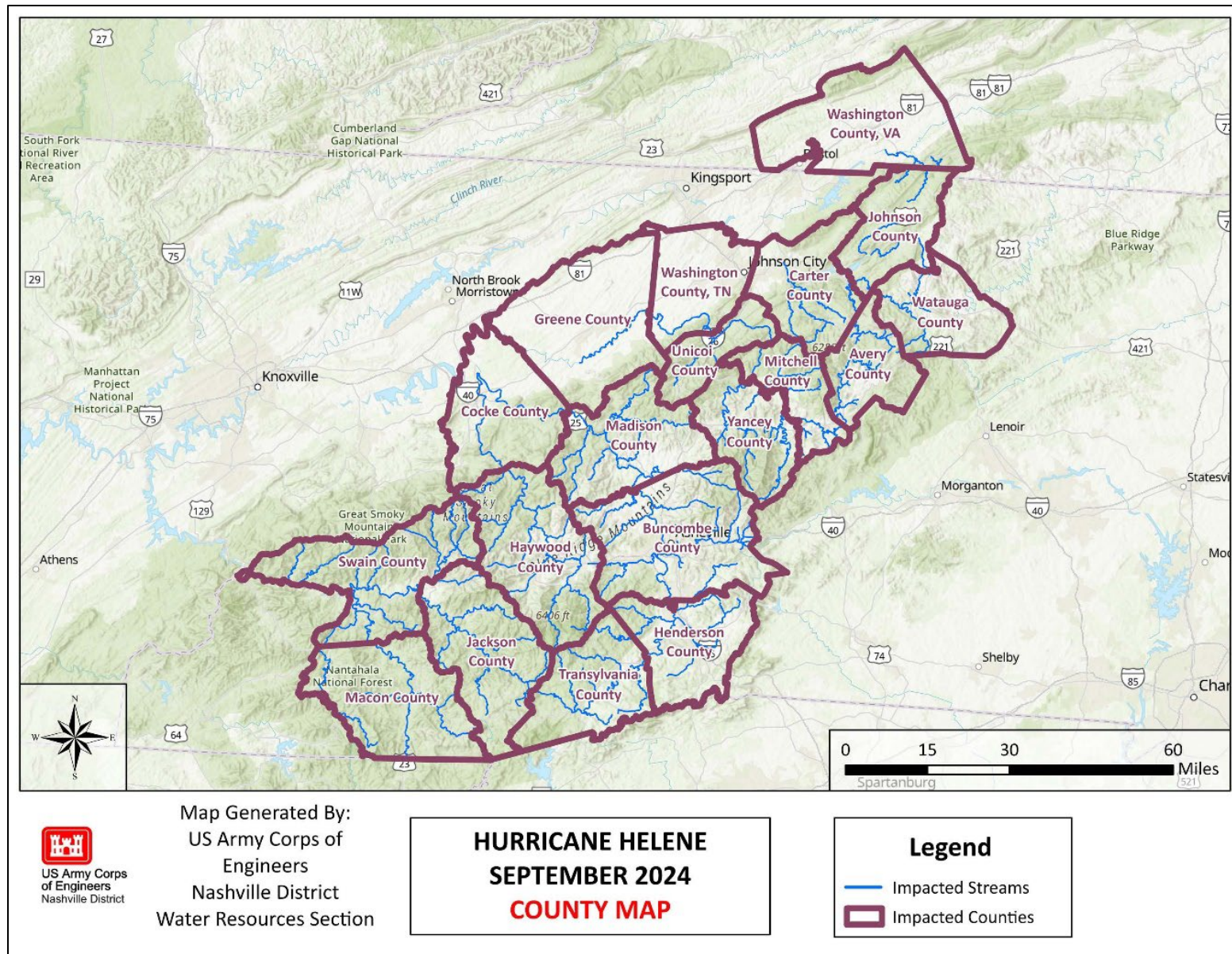


Figure 1.1: Report County Map

2 Coordination

The Hurricane Helene flood response demanded exceptional coordination across all phases of the mission, underscoring the extensive effort involved in ensuring its success. The Mission Manager, acting in a high-level coordination capacity akin to a project manager, was pivotal in overseeing both internal and external efforts at the agency leadership level, ensuring seamless communication and alignment across all parties. Meanwhile, the Mission Specialist, as the engineering expert, focused on detailed technical coordination, offering specialized guidance and support to the USACE team and technical counterparts to address the complexities of the mission. This collaborative effort started several days before the storm made landfall, highlighting the proactive planning and complex execution required to address the challenges presented by the disaster. The sections below demonstrate the external and internal coordination led by the Mission Manager and Mission Specialist throughout the mission.

2.1 Interagency Coordination

Coordination with partner agencies began during the scoping phase of the mission. Per PL 84-99 Class 210 requirements, the technical assistance the USACE provides must be supplemental to State and local actions. Therefore, the first agencies contacted were the Tennessee Valley Authority (TVA), United States Geological Survey (USGS), and the National Weather Service (NWS). These water resource partners are key in flood events since they monitor their gage networks and projects throughout a forecasted event. TVA collected the high-water mark data in Tennessee at their water control structures and downstream of these infrastructure projects along the Tennessee River Basin. The USGS in Tennessee and North Carolina only collected high-water marks at their gage locations, as their main focus was equipment recovery. The USGS in Virginia collected high-water marks in disaster-declared counties within the state of Virginia through a FEMA mission assignment, in addition to HWM collection at their gage locations. The NWS did not collect any high-water mark data, but instead provided other relevant historical data, such as climatological conditions.

Emergency management groups such as FEMA, Tennessee Emergency Management Agency (TEMA), Virginia Department of Emergency Management (VDEM), and North Carolina Emergency Management (NCEM) (also known as the North Carolina Department of Public Safety, or NCDPS) were coordinated with to determine road conditions, recovery efforts, emergency contact information, and high-water mark collection efforts. Only NCEM collected high-water mark data on affected streams, concentrating on gage locations and damage centers. These partners were critical in developing and sharing the public information flyers in addition to data collection efforts. They were able to inform local emergency managers of our arrival and provide support to the field team when needed.

Much of the aforementioned coordination directly affected the LRN team's collection efforts by narrowing the scope of work to avoid duplication and aiding the Nashville District's navigation. Coordination helped identify resource sharing opportunities and gaps in authorities for each agency. Other agency coordination occurred at Silver Jackets meetings, Task Force meetings, and email correspondence. Several other local, state, and private entities have contacted the Nashville District to determine if the data could be useful in their recovery efforts. This report will be made available to our partners and the public.

HURRICANE HELENE – POST FLOOD REPORT

Data from other USACE districts or outside agencies are not included in the scope of this report. This dataset represents only the coverage within the Nashville District area of responsibility and does not encompass the full extent of the hurricane. For additional information on data from other agencies, please contact the respective organizations directly.

2.2 USACE Coordination

The Hurricane Helene flood response required coordination between different sections and districts within the USACE. The Nashville District's (LRN) Hydraulics and Hydrology (H&H) section led the HWM effort, with collaboration and support from the Project Planning Branch, several offices within Engineering and Construction Division, Emergency Management Office, Public Affairs Office, and Safety Office. Furthermore, support for the HWM effort also came from the Memphis District's (MVM) Surveying Section. This section highlights the intra-agency coordination on funding, personnel, equipment, and public outreach.

The mission began with the establishment of a funding source, which, due to the widespread impact of Hurricane Helene, required a large request routed through EM to HQUSACE. Coordination was essential to develop a cost estimate for the effort, which extended from an initial 4–6-week effort in two states to 8 weeks across three states.

Teams were established and deployed based on immediate availability, with each team consisting of three members: a H&H team lead, a representative from the LRN or MVM survey section, and an additional Nashville District team member for data collection. Coordination with the Safety Office and Public Affairs Office (PAO) ensured team members were briefed on hazards and provided appropriate personal protective equipment (PPE), as well as media and trauma response training. Public Outreach was a critical component of the team's success with the support of PAO in developing flyers and pamphlets for the impacted communities. Daily communication between the Mission Manager, field teams, and the EM office was critical, with formal SITREPs provided each afternoon. Coordination with the LRN tribal liaison was necessary before collecting data in the "Qualla Boundary", or territory of the Eastern Band of the Cherokee Indians, ensuring cultural sensitivity training and tribal permission.

Finally, coordination was extended for equipment needs ensuring the logistical success of the mission. Transportation arrangements were coordinated with multiple sections ensuring that sufficient vehicles, including four-wheel-drive trucks for rugged terrain, and travel reservations were handled appropriately and timely. HWM equipment, including field tools and maps, was supplied by the H&H section with new purchases or repurposed function, while all survey equipment was provided by the respective surveying sections.

3 Hurricane Helene Event

3.1 Historic Storms and Floods

Due to its high terrain and river valleys, flooding is common in western North Carolina. Notable flooding and devastation came from Hurricane Frances and Hurricane Ivan, both occurring in 2004. These events triggered landslides and extreme flooding in western North Carolina; however, the most notable flood in the area prior to Hurricane Helene was the “Great Flood of 1916.” This event was the result of two consecutive hurricanes in July of 1916, with one originating in the Gulf and the other forming in the Atlantic. The first storm made landfall on the Gulf Coast of Alabama then slowly moved north where it produced heavy rainfall over the mountains of North Carolina before it weakened and dissipated over southeast Tennessee. While the first storm did not produce flooding, it did leave the area saturated with streams running high when the second hurricane made landfall on the coast of South Carolina. This storm moved west and reached the North Carolina mountains as a tropical storm, producing record widespread rainfall (National Weather Service, n.d.) The first storm caused the French Broad River to raise to 8.8 feet, 4.8 feet above flood stage. While it did recede to 4.0 feet before the second event hit, the second hurricane brought over 10 inches of rain to the watershed, and the French Broad River crested at 23.1 feet at the gage in Asheville, NC (North Carolina Geological Survey and UNC Asheville's NEMAC, n.d.). The flood brought widespread devastation, historic flooding, landslides, and resulted in numerous fatalities.

Another notable flood in the area, specifically in Haywood County, occurred in August 2021 as a result of Tropical Storm Fred. The majority of the heavy rainfall fell in the East Fork Pigeon River, upstream of Cruso, NC. Peak stage on the East Fork Pigeon River exceeded its previous flood-of-record by 3 feet, rising to 16.03 feet during this event. This storm brought catastrophic flash flooding and major landslides to the region, damaging and destroying multiple homes and resulting in six fatalities in the area. As the water continued to flow downstream, major flooding and damage was also seen on the Pigeon River in Canton, NC.

Similarly, East Tennessee has also experienced extreme flooding. The most notable flood on record for the area, until Hurricane Helene, was the flood of 1901. Multiple rivers, including the Doe River and Watauga River, experienced such severe flooding that it was reported to have changed their courses in some areas. The flood was a result of intense rainfall from late February into the first of March.

The NWS publishes historic stage data at its streamflow gages for the United States. The NWS provide flood categories at many gages ranging from “action” to “major” flooding levels based on set stage heights. The “major” flood category represents *“extensive inundation of structures and roads. Significant evacuations of people and/or transfer of property to higher elevations”*. This category was used to get an idea for major historic flooding in the impacted watersheds from Hurricane Helene. Table 3.1 below lists the NWS gages gathered in this report, along with the period of record, and how many times the major flood stage has been surpassed in history. The gages can also be seen in Figure 3.1 which provides a map of the gages in the watersheds.

HURRICANE HELENE – POST FLOOD REPORT

Table 3.1: NWS Gage Table – Major Floods

Number	NWS Gage ID	NWS Gage Name	Stream	Watershed	Number of Major Floods	Period of Record
1	NEPT1	Pigeon River at Newport, TN	Pigeon River	Pigeon River	16	1900 - Present
2	CTPN7	Pigeon River at Canton, NC	Pigeon River	Pigeon River	5	1985 - Present
3	NWPT1	French Broad River near Newport, TN	French Broad River	French Broad River	4	1867 - Present
4	MARN7	French Broad River near Marshall, NC	French Broad River	French Broad River	11	1916 - Present
5	AVLN7	French Broad River at Asheville, NC	French Broad River	French Broad River	3	1896 - Present
6	FLCN7	French Broad River at Fletcher, NC	French Broad River	French Broad River	2	2001 - Present
7	BLAN7	French Broad River at Blantyre, NC	French Broad River	French Broad River	7	1875 - Present
8	BLTN7	Swannanoa River at Biltmore, NC	Swannanoa River	French Broad River	8	1876 - Present
9	EMBT1	Nolichucky River at Embreeville, TN	Nolichucky River	Nolichucky River	3	1901 - Present
10	EZBT1	Watauga River at Elizabethton, TN	Watauga River	Watauga River	6	1867 - Present
11	SGWN7	Watauga River near Sugar Grove, NC	Watauga River	Watauga River	5	1916 - Present
12	DOET1	Doe River at Elizabethton, TN	Doe River	Watauga River	3	1901 - Present
13	TKSN7	Tuckasegee River at Bryson City, NC	Tuckasegee River	Tuckasegee River	2	1840 - Present

HURRICANE HELENE – POST FLOOD REPORT

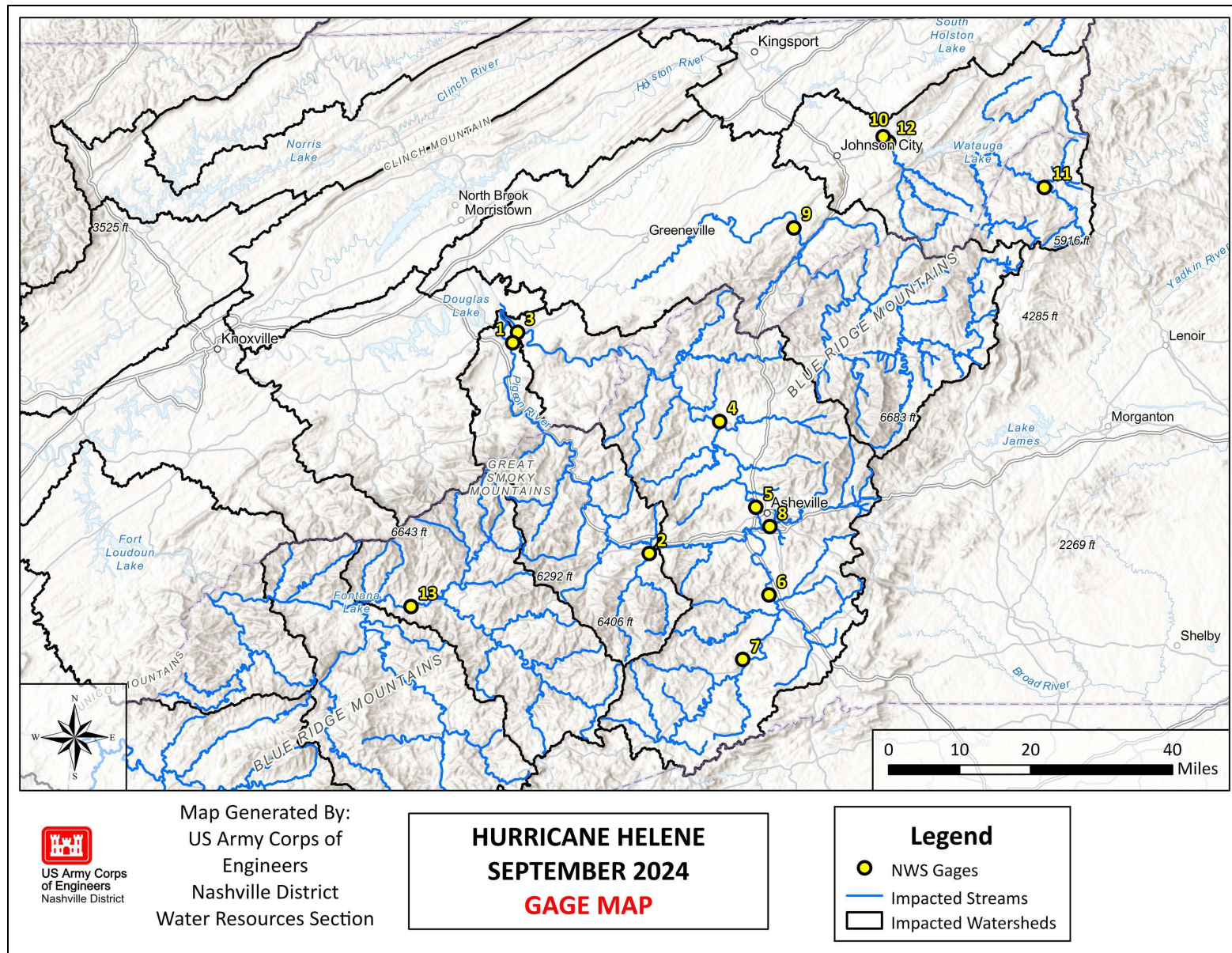


Figure 3.1: NWS Gages Map

HURRICANE HELENE – POST FLOOD REPORT

3.2 Antecedent Conditions

The weather conditions leading up to the storm, including pre-hurricane rainfall on September 25 and 26, aided in the extreme flooding of Hurricane Helene. According to the National Oceanic and Atmospheric Administration (NOAA), Hurricane Helene contained all of the ideal and specific meteorological conditions to produce heavy tropical moisture in the inland areas of the United States. Extended periods of rainfall increase soil moisture and stream flows, thereby increasing the potential for runoff. Some areas in the mountainous region of North Carolina, Tennessee, and Virginia received 6 to 10 inches of rainfall. According to the NWS, heavy rainfall started late September 25 and continued through September 26 when the front stalled across the region. Figure 3.2 below shows the 48-hour cumulative rainfall from the NOAA Multi-Radar Multi-Sensor (MRMS) Quantitative Precipitation Estimate (QPE) rainfall grids from September 24, 1200, EST – September 26, 1200, EST. The rainfall grid was separated into five levels based on the total rainfall over the period.

HURRICANE HELENE – POST FLOOD REPORT

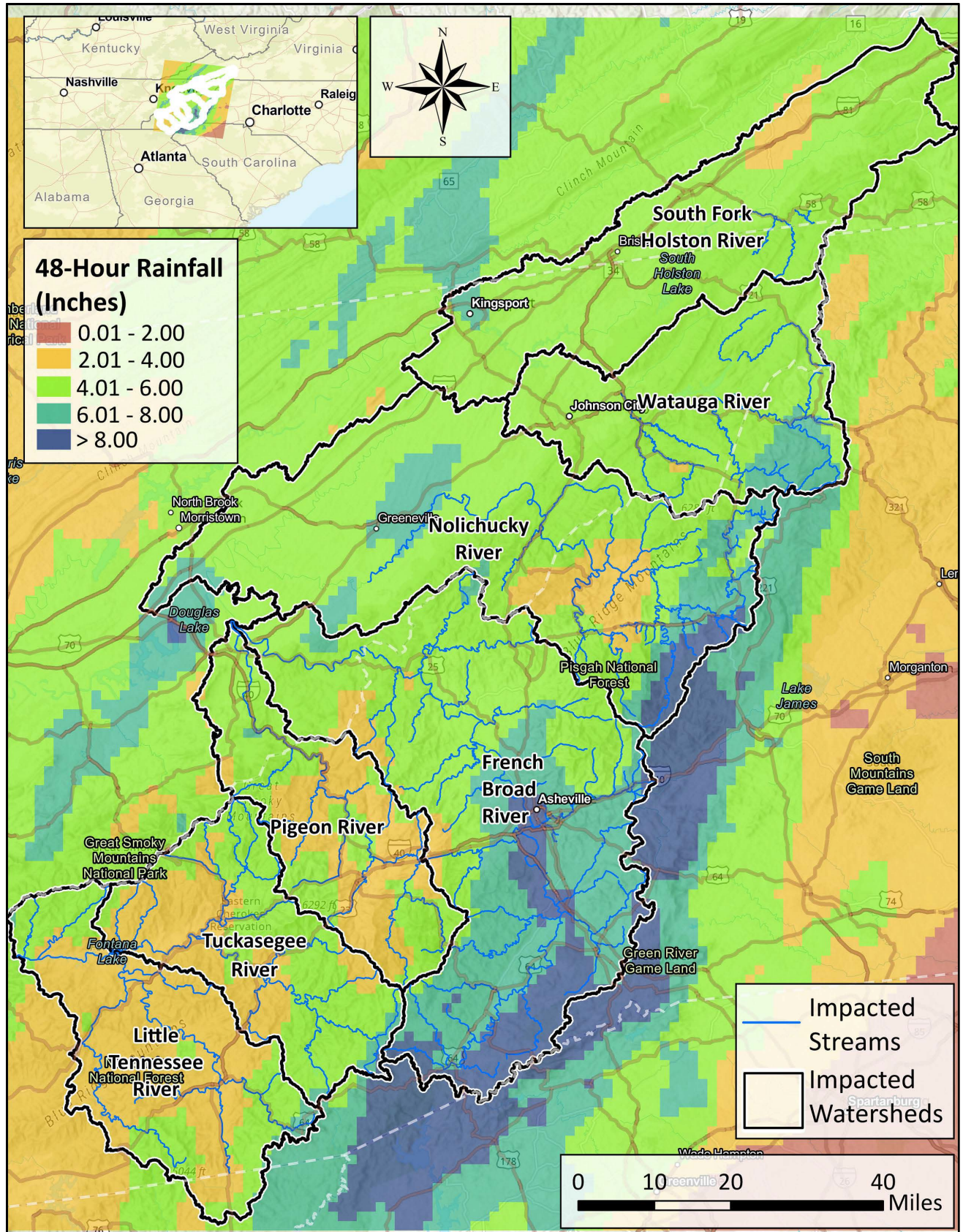


Figure 3.2: Antecedent Storm – September 24, 1200, EST – September 26, 1200, EST – Rainfall Totals

HURRICANE HELENE – POST FLOOD REPORT

The heavy rainfall preceding Hurricane Helene's arrival contributed to adverse hydrologic conditions. The region's soil was already saturated, leaving it unable to absorb any significant amount of additional precipitation. As a result, the subsequent rainfall from Hurricane Helene rapidly translated into runoff, inducing flash flooding and amplifying flows in the watersheds. Furthermore, the antecedent storms had already pushed stream stages to elevated levels, as evident in the stage hydrographs shown in Figure 3.3 – Figure 3.6. By 1500 EST on September 26, 2024, USGS gages in the region recorded elevated stages, with most streams either at or near flood stage by the evening. When the intense rainfall from Hurricane Helene hit on September 27, the already saturated watersheds responded with historic flooding, leading to devastating impacts to the communities along the impacted rivers.

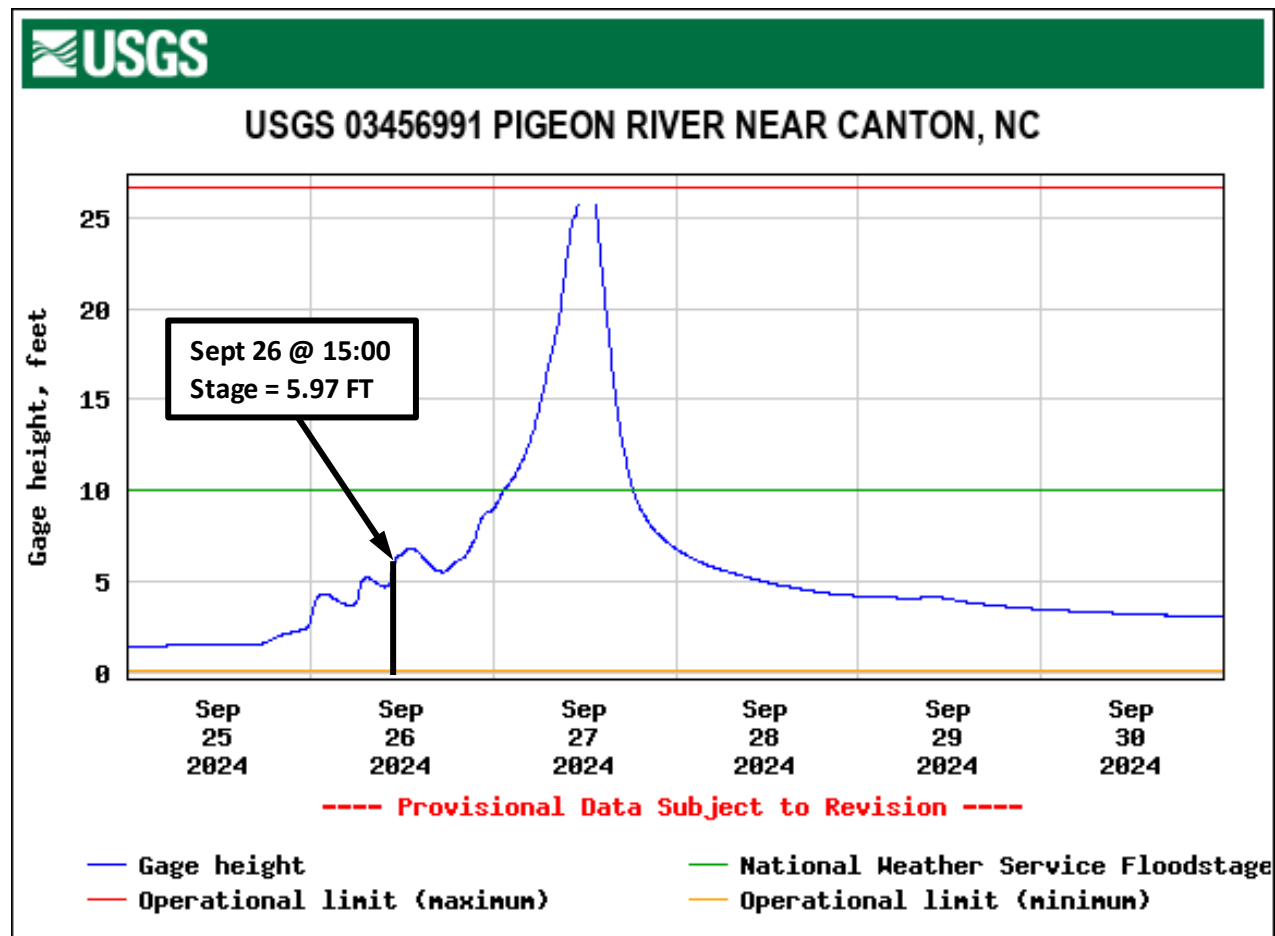


Figure 3.3: USGS Gage 03456991 Pigeon River near Canton, NC – Stage Hydrograph – Antecedent Stage

HURRICANE HELENE – POST FLOOD REPORT

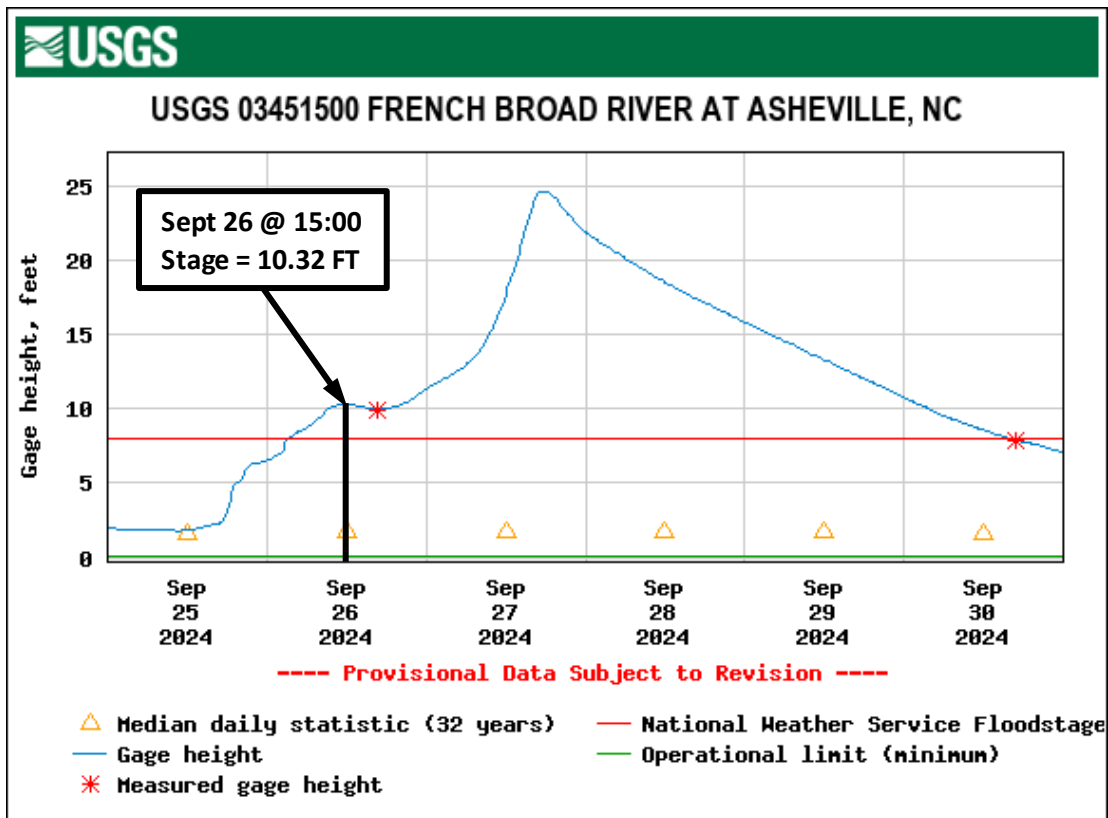


Figure 3.4: USGS Gage 03451500 French Broad River at Asheville, NC – Stage Hydrograph – Antecedent Stage

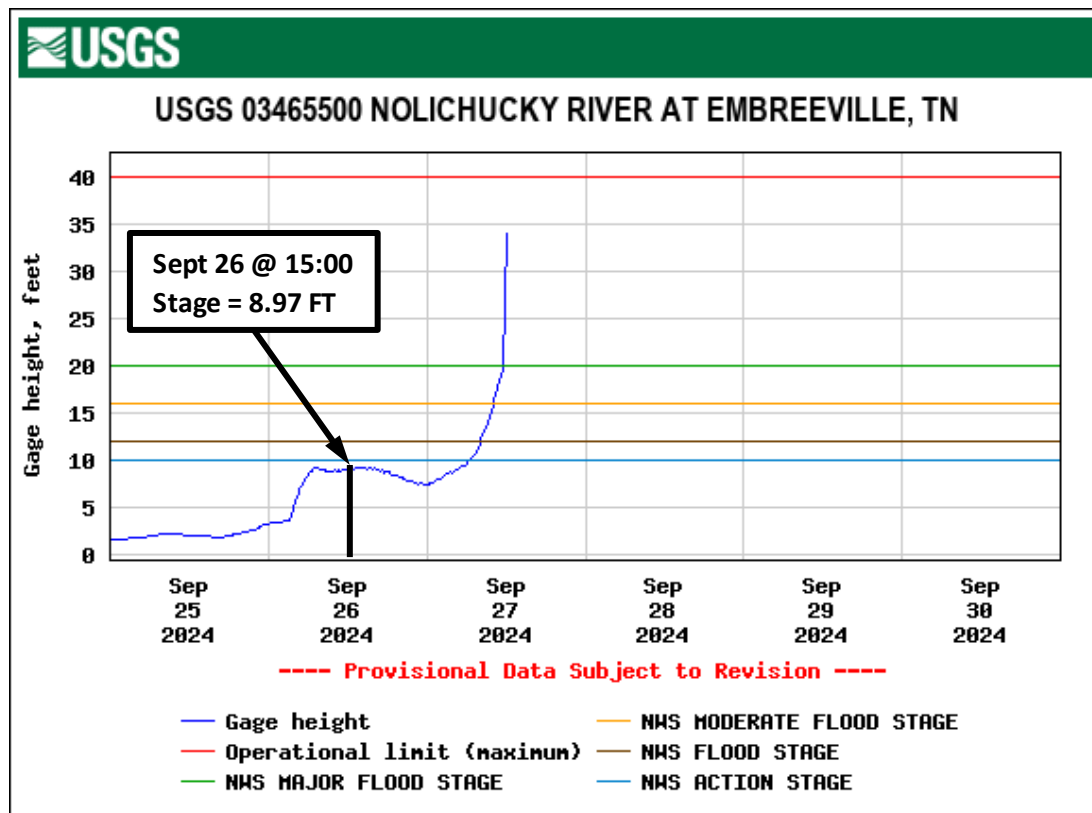


Figure 3.5: USGS Gage 03465500 Nolichucky River at Embreeville, TN – Stage Hydrograph – Antecedent Stage

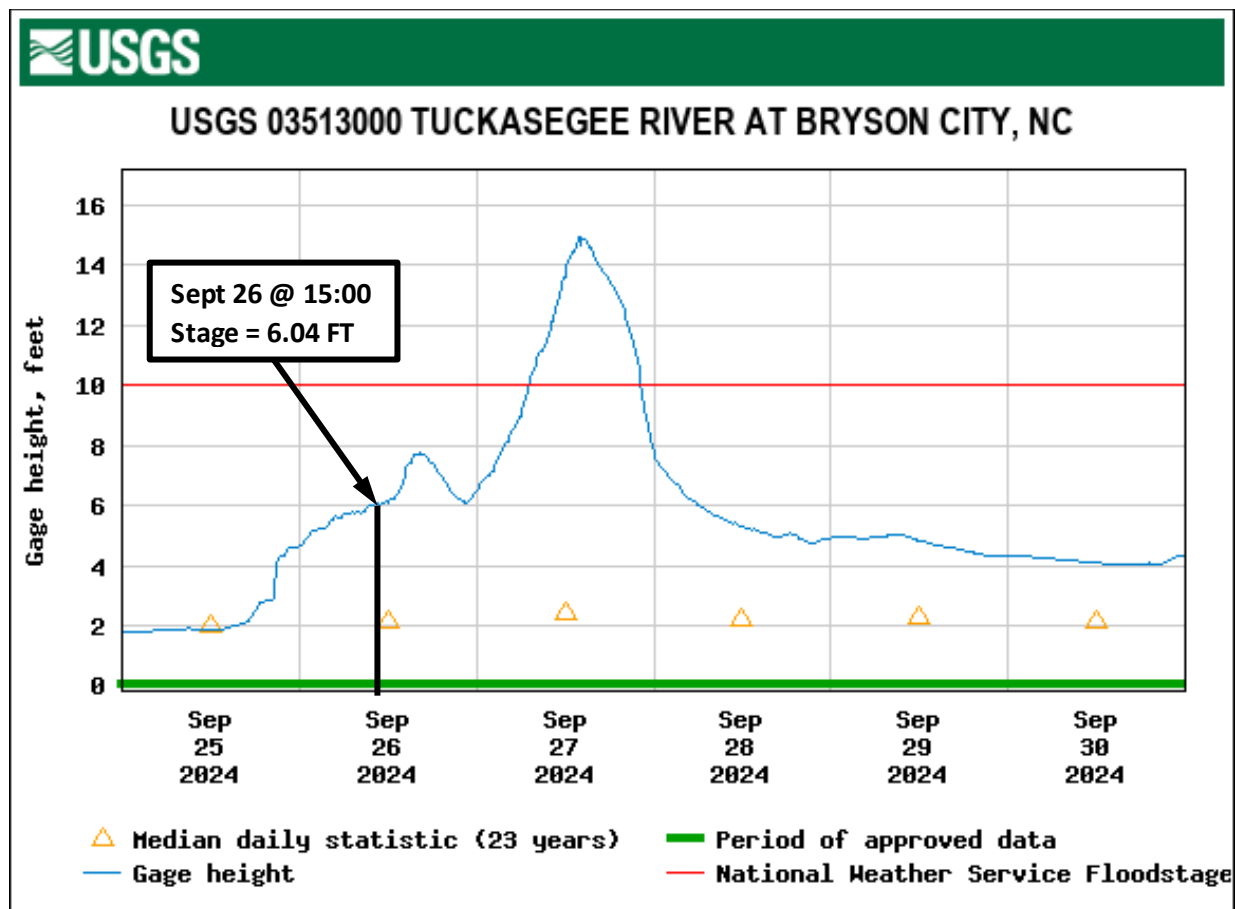


Figure 3.6: USGS Gage 03513000 Tuckasegee River at Bryson City, NC – Stage Hydrograph – Antecedent Stage

3.3 Hurricane Helene Rainfall

Just before midnight on September 27, Hurricane Helene made landfall in the Florida’s Big Bend region. The outer bands of rainfall spread into Georgia and the Carolinas. The storm’s most intense rainfall occurred on the morning of September 27, as Helene interacted with the mountainous terrain of western North Carolina and eastern Tennessee before moving away from the region in the afternoon.

By the evening of September 27 as the storm abated, the 3-day rainfall totals revealed a stark contrast between western North Carolina and eastern Tennessee. Western North Carolina bore the brunt of the storm, with widespread estimates of 12-20 inches of rainfall and isolated areas receiving over 20 inches. Although eastern Tennessee received relatively less precipitation, with totals ranging from 6-10 inches, the region was not spared from the devastating effects of the storm. Many of the impacted stream’s flow from North Carolina into Tennessee, causing major flooding in both states. This extreme rainfall over saturated soils resulted in catastrophic and historic flooding across western North Carolina and eastern Tennessee.

Figure 3.7 provides a visual representation of the total 3-day rainfall accumulation in the southern Appalachians, as measured by NOAA MRMS QPE. The data is categorized into five distinct shades, with the darkest blue shade indicating the highest rainfall totals at greater than 15 inches of rainfall. Notably, this intense rainfall zone encompasses the upper reaches of the French Broad, Nolichucky, and Watauga watersheds, highlighting areas of significant precipitation.

HURRICANE HELENE – POST FLOOD REPORT

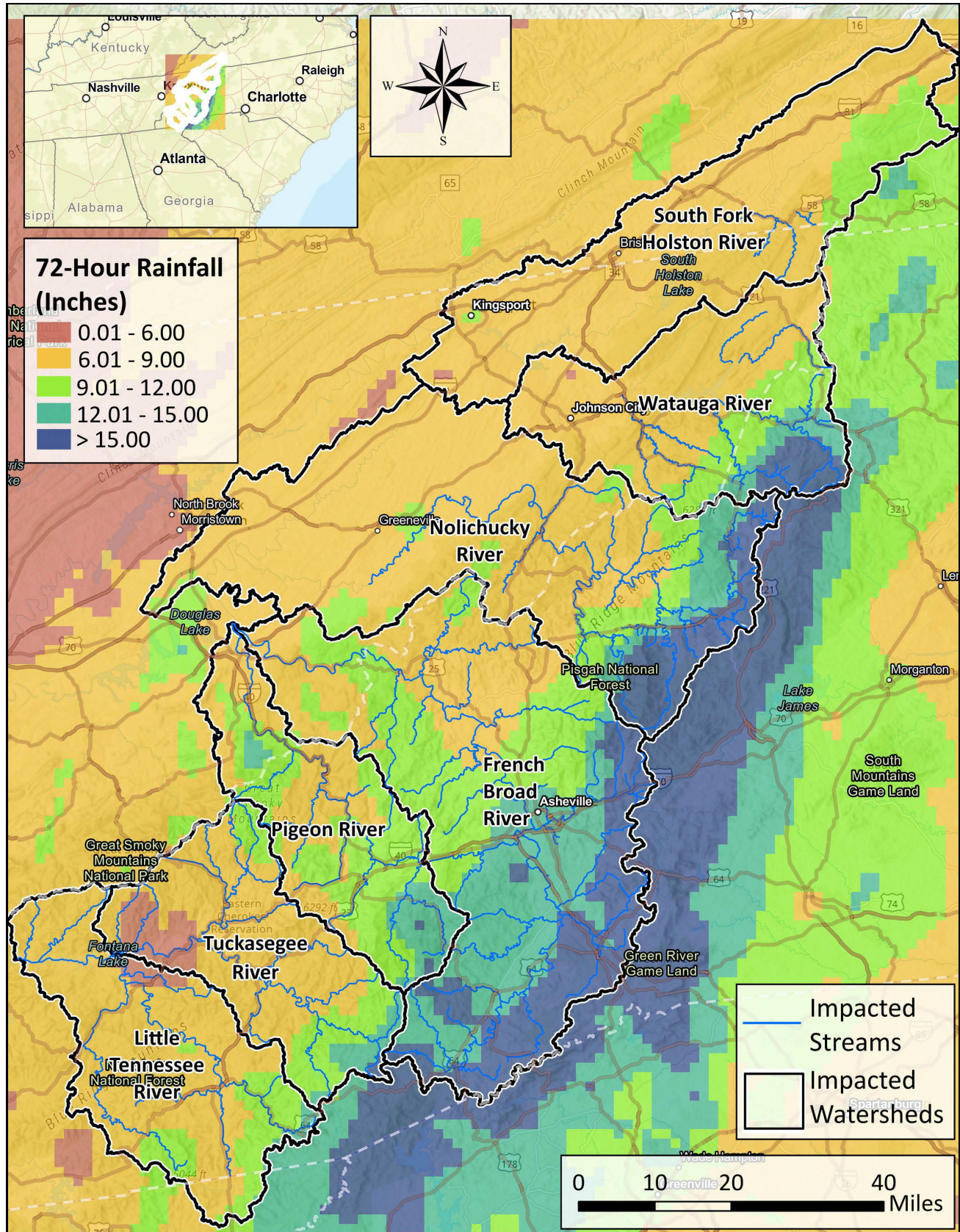


Figure 3.7: Hurricane Helene Rainfall—September 24, 1200, EST–September 27, 1200, EST – 3-Day Rainfall Totals

HURRICANE HELENE – POST FLOOD REPORT

The 3-day rainfall totals for the Hurricane Helene rainfall event were compared to the NOAA Atlas 14 precipitation frequency estimates by the NWS. Atlas 14 is a study of historical rainfall data done to help understand flood risk, rainfall intensities, and precipitation frequency estimates. It was completed in 2006 and is the official U.S. Government's source of precipitation frequency estimates (National Oceanic and Atmospheric Administration, n.d.). The graphic in Figure 3.8, developed by NOAA, displays the annual exceedance probability of the rainfall event. The black shaded region in Asheville and over a large area in North Carolina is the 0.001 (1/1000, 1000-year) annual exceedance probability precipitation depth. This indicates that the 3-day rainfall totals in this region were so extreme that the statistical chances of them being exceeded in an any given year were 1 in 1,000 according to NOAA Atlas 14. This figure highlights the annual exceedance probability of the event in the context of NOAA Atlas 14 precipitation depths.

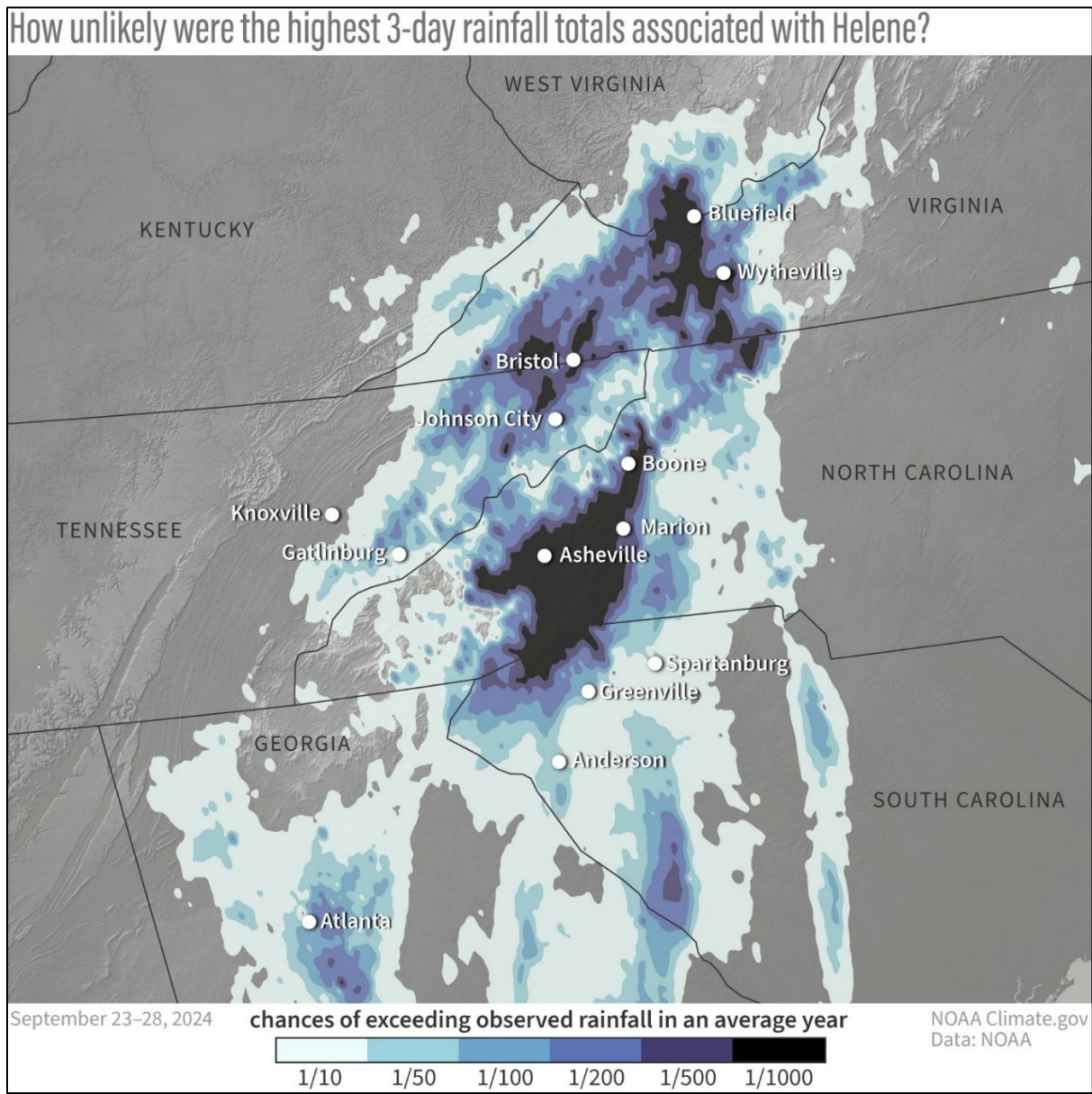


Figure 3.8: NOAA Hurricane Helene 3-day Rainfall Annual Exceedance Probabilities (*National Oceanic and Atmospheric Administration, 2024*)

HURRICANE HELENE – POST FLOOD REPORT

The Community Collaborative Rain, Hail, and Snow Network (CoCoRaHS) provides critical precipitation data through a community-driven effort, where volunteers collect and map rainfall measurements that are made publicly available on their interactive website. Notably, this network has extensive coverage in the regions of North Carolina and Tennessee affected by Hurricane Helene. As illustrated in Figure 3.9, the 3-day rainfall accumulation data shows significant precipitation totals, with areas around Asheville, NC (indicated by red icons) experiencing rainfall amounts exceeding 12 inches over the 3-day period.

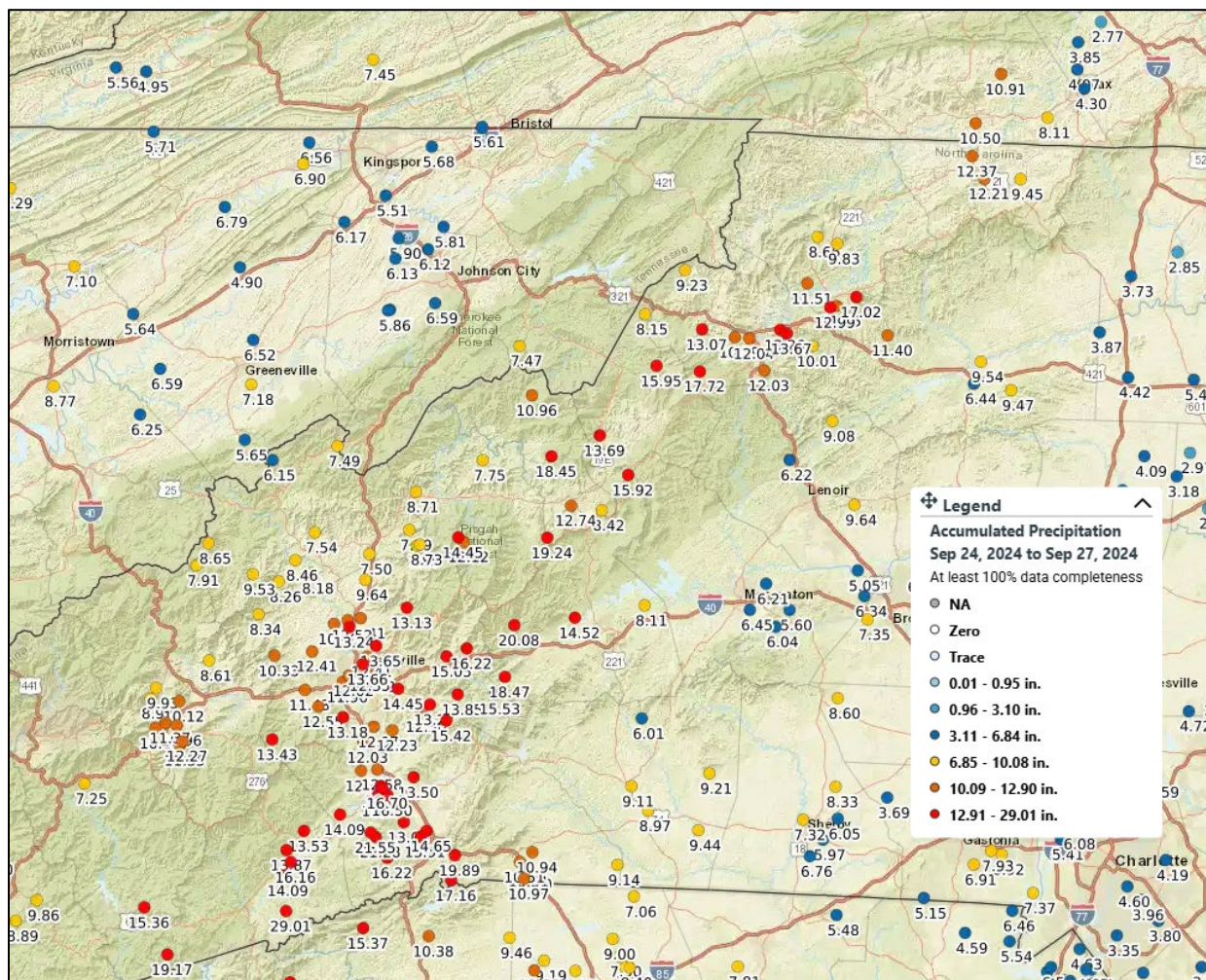


Figure 3.9: CoCoRaHS 3-day Hurricane Helene Rainfall Totals (*Community Collaborative Rain, Hail, and Snow Network, 2024*)

To analyze the temporal pattern and intensity of Hurricane Helene in the southern Appalachians, cumulative precipitation graphs (Figure 3.10 - Figure 3.15) were compiled from USGS gages. These graphs depict the storm's progression from September 24 to September 28, with the blue line representing cumulative precipitation over time. A summary of the precipitation totals, organized by location, is provided in Table 3.2 along with the corresponding NOAA Atlas 14 annual exceedance probabilities which contextualizes the rarity of the observed precipitation totals.

HURRICANE HELENE – POST FLOOD REPORT

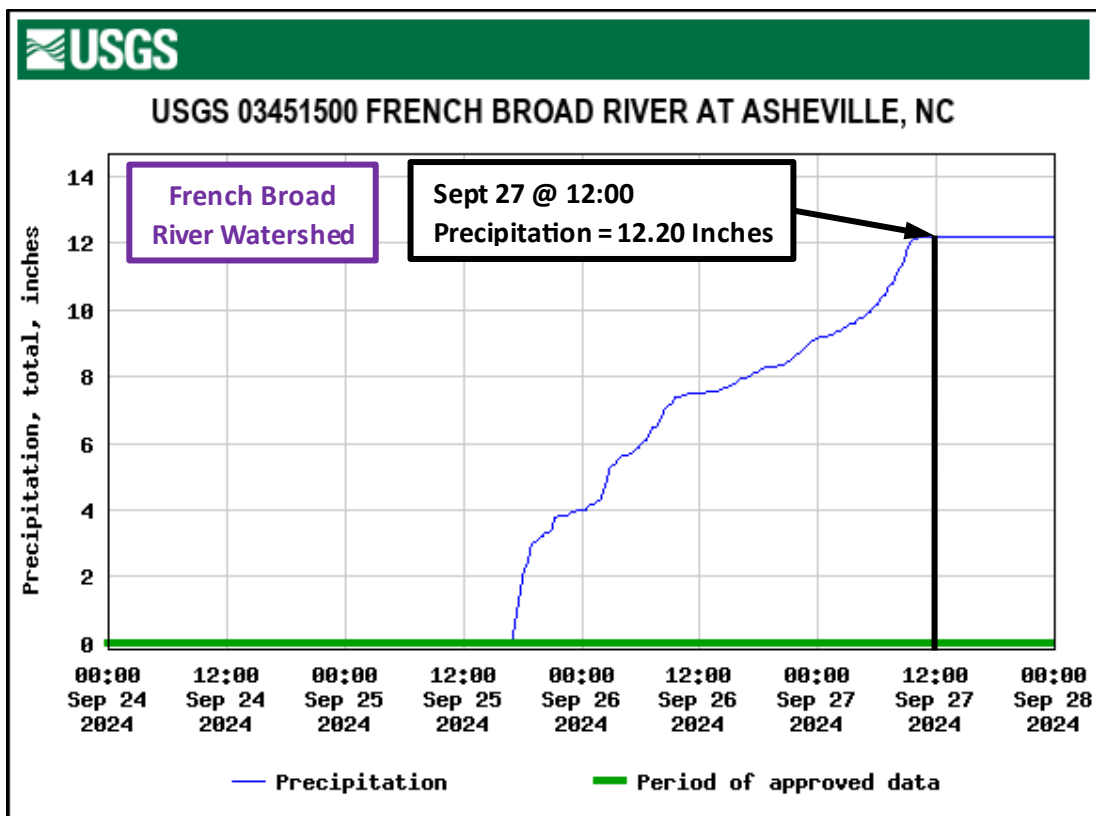


Figure 3.10: USGS Gage 03451500 French Broad River at Asheville, NC - 3-day Precipitation Total

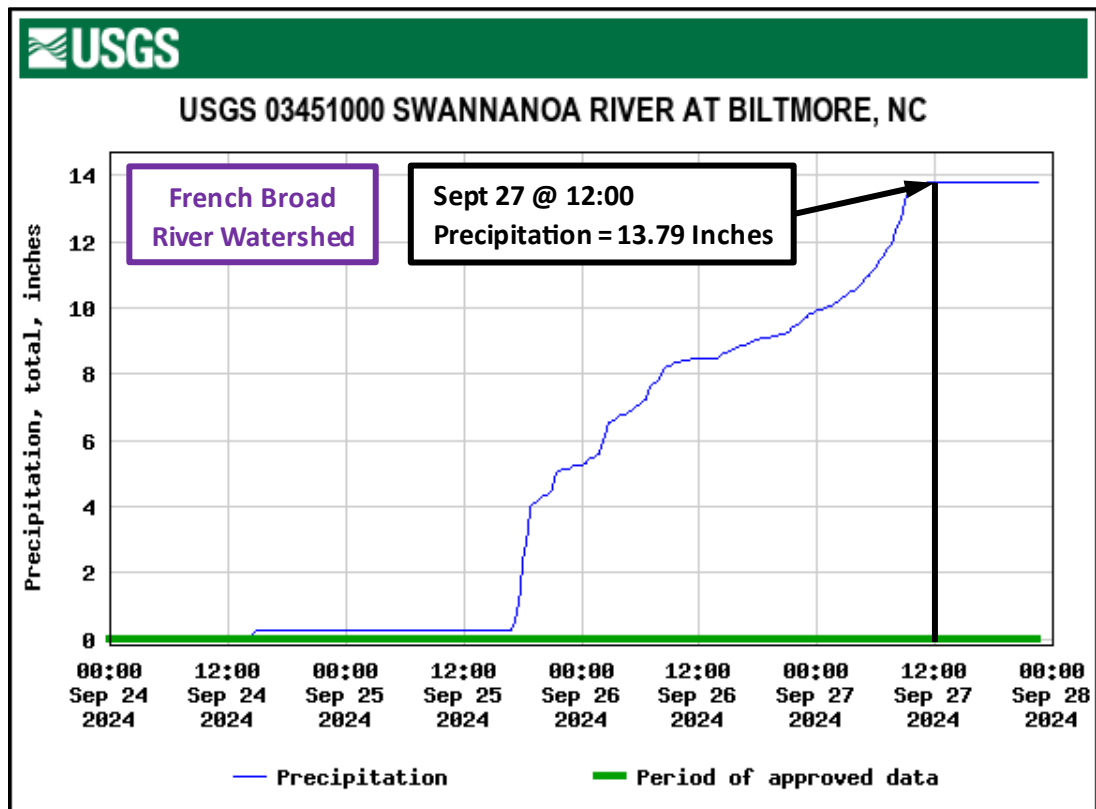


Figure 3.11: USGS Gage 03451000 Swannanoa River at Biltmore, NC - 3-day Precipitation Total

HURRICANE HELENE – POST FLOOD REPORT

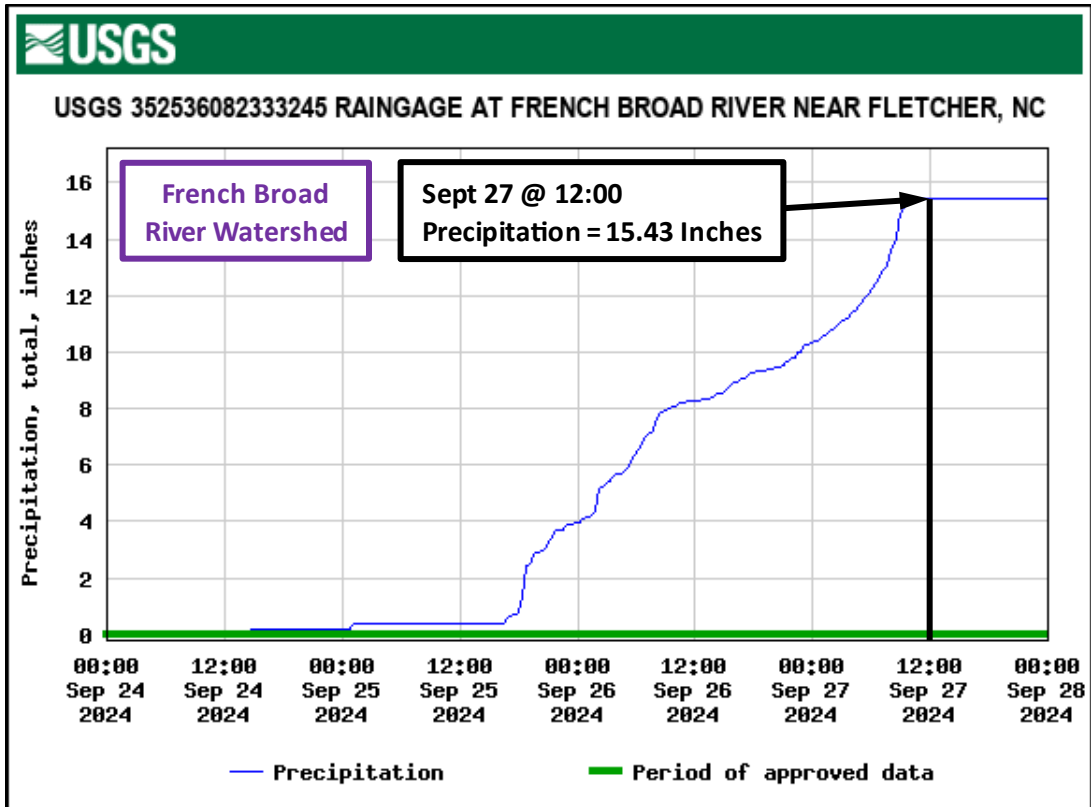


Figure 3.12: USGS Rain Gage French Broad River near Fletcher, NC - 3-day precipitation total

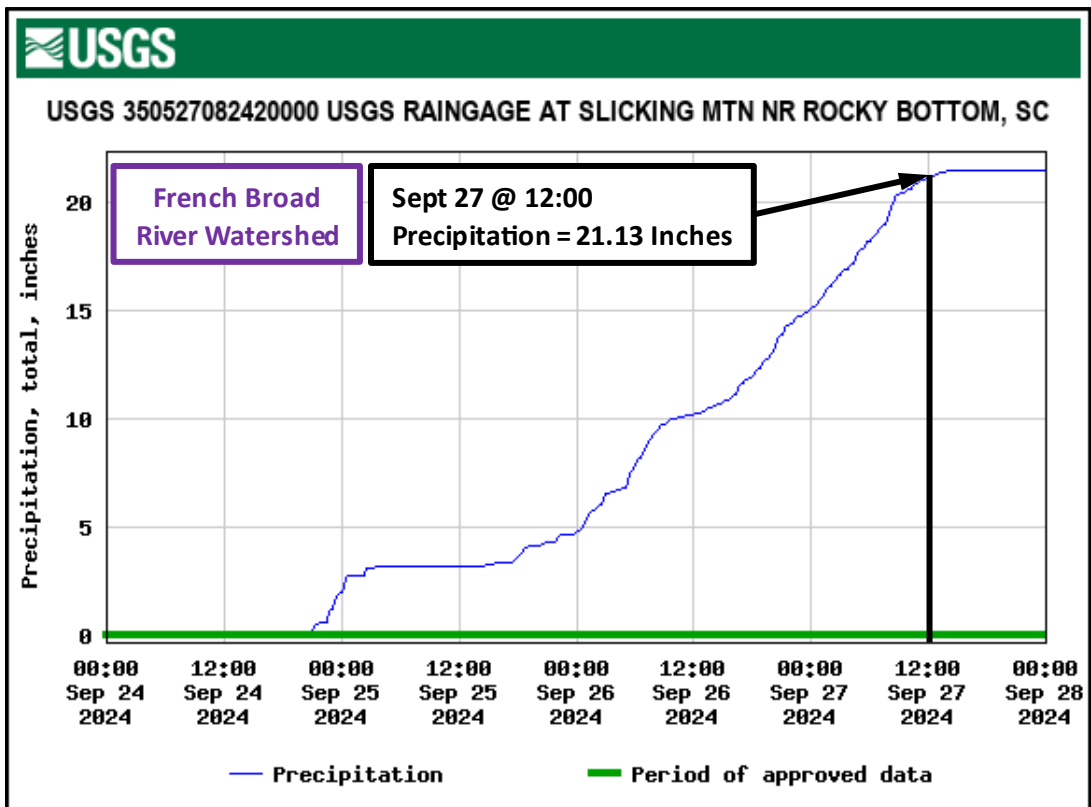


Figure 3.13: USGS Rain Gage Slicking Mtn near Rocky Bottom, SC - 3-day precipitation total

HURRICANE HELENE – POST FLOOD REPORT

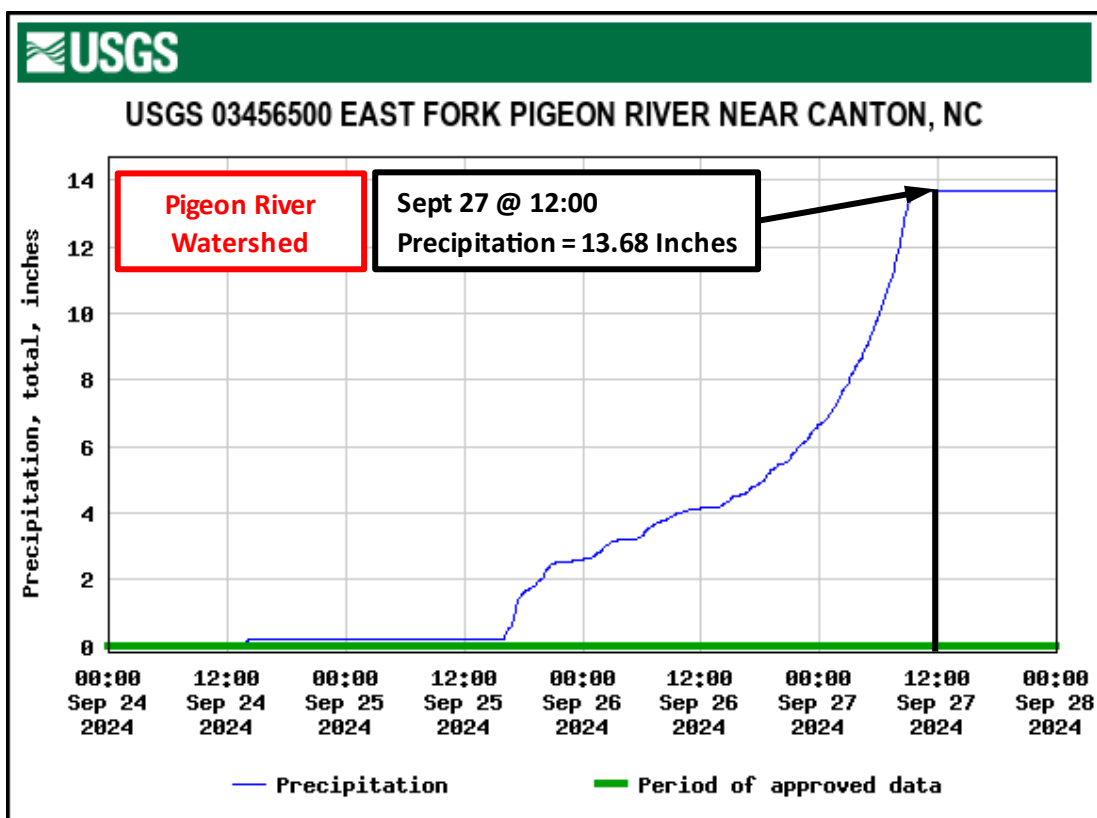


Figure 3.14: USGS Gage 03456500 East Fork Pigeon River near Canton, NC - 3-day precipitation total

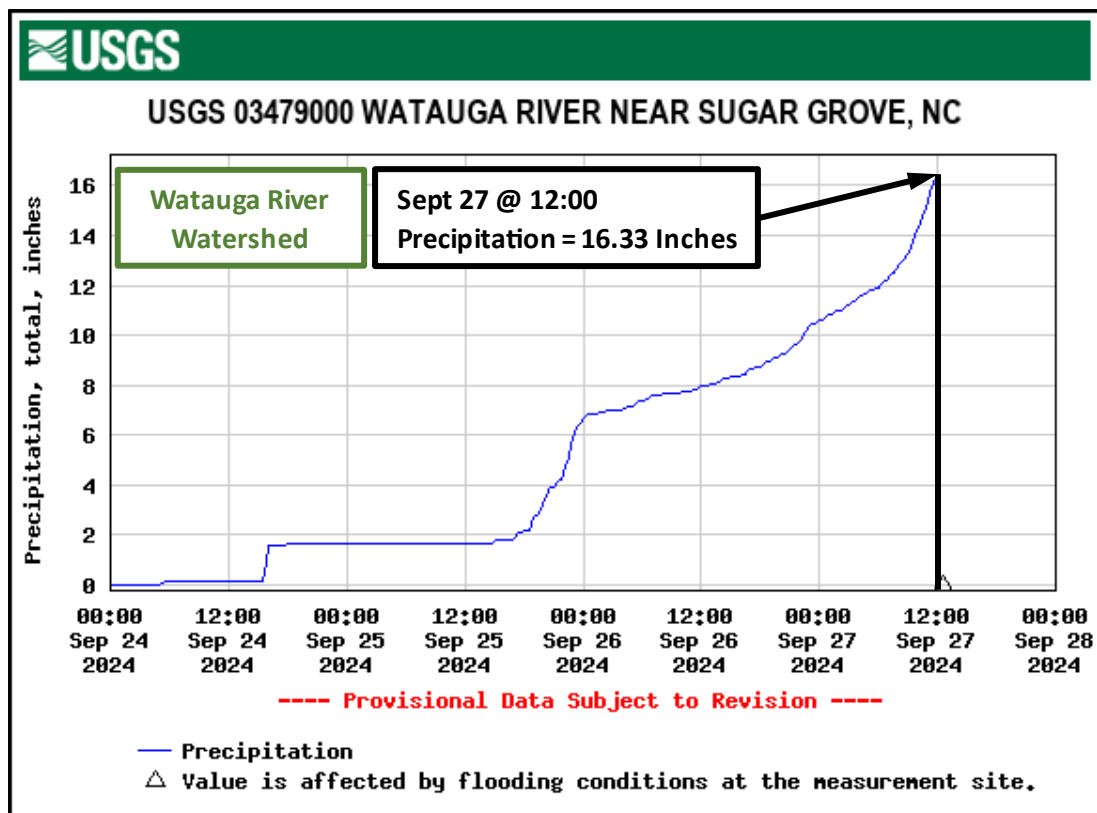


Figure 3.15: USGS Gage 03479000 Watauga River near Sugar Grove, NC - 3-day precipitation total

HURRICANE HELENE – POST FLOOD REPORT

Table 3.2: Hurricane Helene – Precipitation Totals

Watershed	Rainfall Gage Operator	Gage ID	Location	3-Day Rainfall Total, Sept 24 – Sept 27 (Inches)	NOAA Atlas 14 3-Day Annual Exceedance Probability
French Broad River	USGS	3451500	Asheville, NC	12.2	> 1/1000
French Broad River	USGS	3451000	Biltmore, NC	13.8	> 1/1000
French Broad River	USGS	Rain Gage	Fletcher, NC	15.4	> 1/1000
French Broad River	USGS	Rain Gage	Rocky Bottom, SC	21.1	> 1/1000
French Broad River	CoCoRaHS	NC-MS-29	Marshall, NC	8.5	> 1/1000
Pigeon River	USGS	3456500	Canton, NC	13.7	> 1/1000
Pigeon River	CoCoRaHS	NC-HW-72	Waynesville, NC	11.4	> 1/1000
Nolichucky River	CoCoRaHS	NC-ML-15	Buladean, NC	11.0	> 1/200
Nolichucky River	CoCoRaHS	NC-ML-2	Bakersville, NC	18.5	> 1/1000
Nolichucky River	CoCoRaHS	NC-ML-10	Spruce Pine, NC	19.2	> 1/1000
Watauga River	USGS	3479000	Sugar Grove, NC	16.3	> 1/1000
Watauga River	CoCoRaHS	NC-AV-13	Banner Elk, NC	16.0	> 1/500
Watauga River	CoCoRaHS	NC-WT-51	Vilas, NC	13.1	> 1/200
Tuckasegee River	CoCoRaHS	NC-JC-25	Sylva, NC	7.3	> 1/50
Tuckasegee River	CoCoRaHS	NC-JC-22	Cashiers, NC	15.4	> 1/100
Tuckasegee River	CoCoRaHS	NC-TR-29	Sapphire, NC	19.2	> 1/500

3.4 Hurricane Helene Peak Stage and Flow Data

Steep topography, saturated soils from previous rainfall, and lack of storage in the mountainous valleys helped contribute to the magnitude of this event. Hurricane Helene was a record-breaking event for both the western North Carolina and eastern Tennessee regions, with the Pigeon, French Broad, and Swannanoa Rivers all exceeding previous stage records.

The following sections help summarize and compare historical peaks to the peak stage and flow data observed during this event at specific, major gage locations throughout the watersheds of interest. The gages used in these sections are numbered in Figure 3.16 based on the gage map number listed in the stage tables listed in the following sections.

HURRICANE HELENE – POST FLOOD REPORT

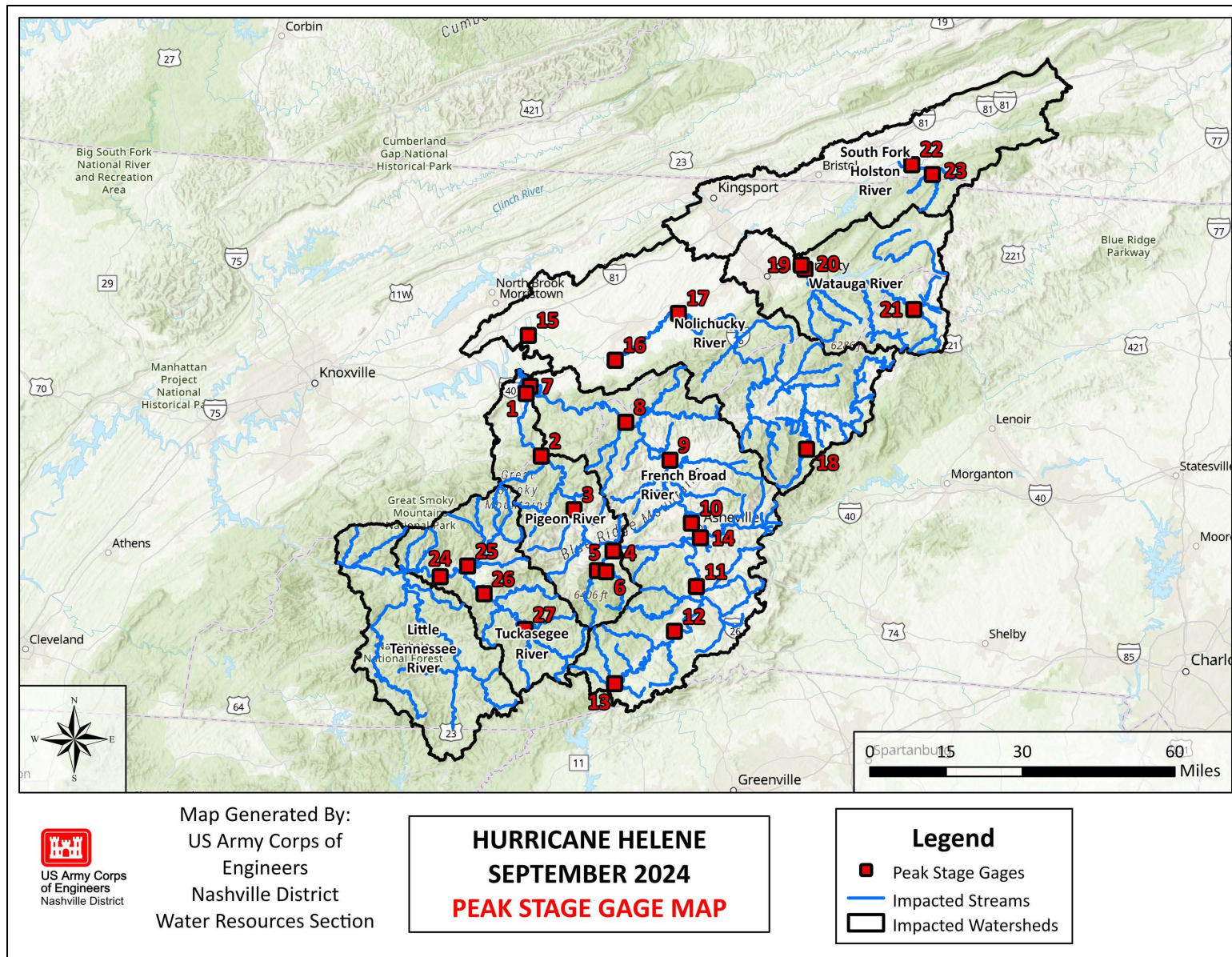


Figure 3.16: Hurricane Helene - Peak Stage Gage Map

HURRICANE HELENE – POST FLOOD REPORT

3.4.1 Pigeon River Watershed

The Pigeon River saw previous records exceeded by 2 to 8 feet at various gage locations on the river. As seen in Table 3.3 and Figure 3.18 below, five out of six of the USGS gages analyzed in the Pigeon River watershed exceeded their previous stage records, and all six gages exceeded the FEMA 100-year estimated stage. The most downstream USGS gage identified on the Pigeon River, USGS 03461500 Pigeon River at Newport, TN, broke its previous stage record set in 1902 at 21.4 feet, by 8.33 feet, peaking at 29.73 feet, displayed in Figure 3.17. This illustrates the point made in Section 3.3, that while Tennessee did not experience the same intensity of rainfall, the flooding effects felt downstream were just as severe.

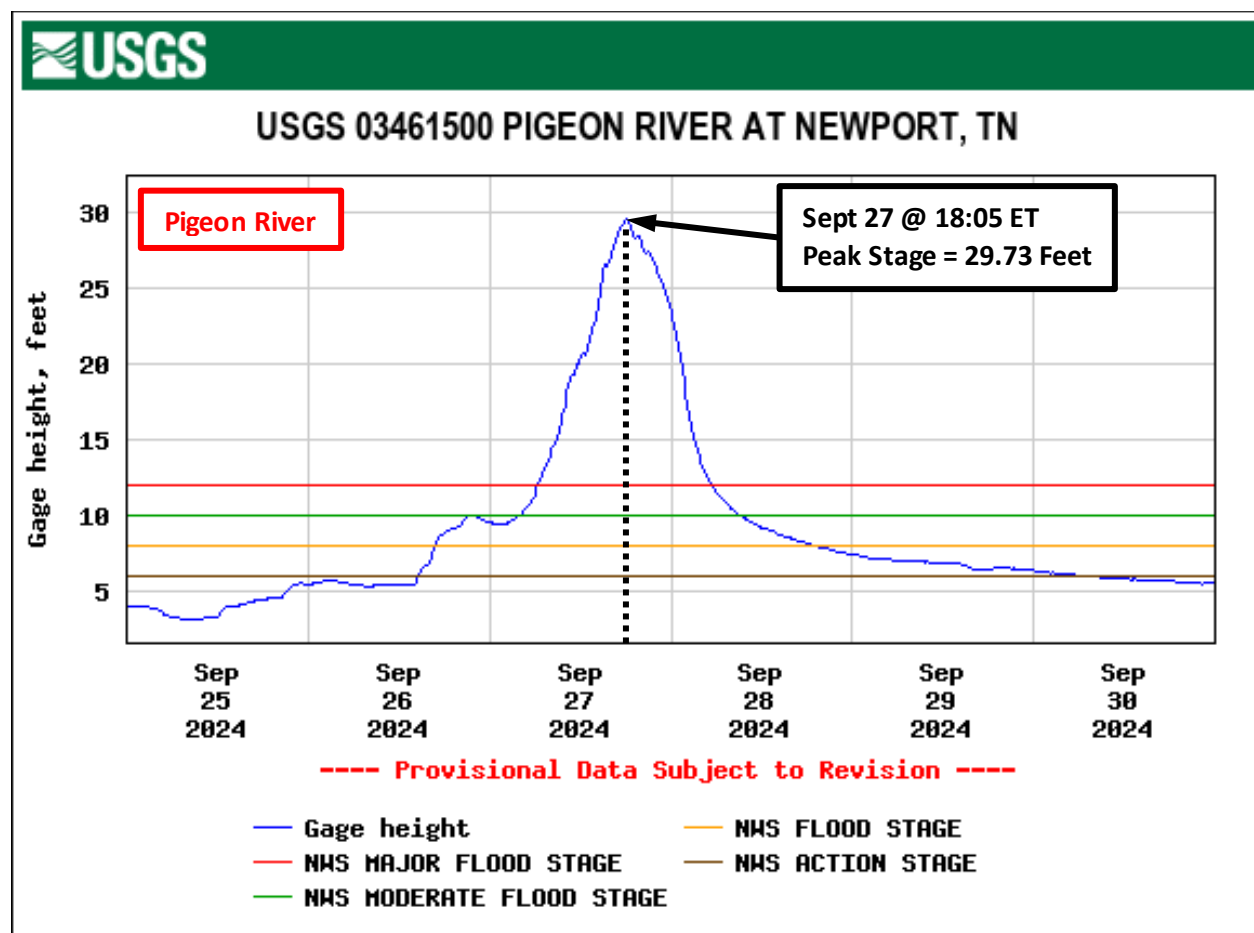


Figure 3.17: USGS Gage 03461500 Pigeon River at Newport TN – Stage Hydrograph

HURRICANE HELENE – POST FLOOD REPORT

Table 3.3: Pigeon River Watershed - Gage Data

PIGEON RIVER WATERSHED – STREAMFLOW GAGE TABLE														
Gage Map Number	NWS Gage ID	USGS Gage ID	Gage Name	Stream	Watershed	Drainage Area (sq. miles)	Period of Record	Historic Peak Stage and Discharge before Hurricane Helene Flood			Hurricane Helene Peak Stage and Discharge			Helene New Historic Peak?
								Peak Stage (Feet)	Peak Discharge (CFS)	Date of Peak	Peak Stage (Feet)	Peak Discharge (CFS)	Date and Time of Peaks (ET)	
1	NEPT1	3461500	Pigeon River at Newport, TN	Pigeon River	Pigeon River	666.0	1900 - Present	21.4	50,000	2/28/1902	29.73	N/A	9/27/2024 18:05	Yes
2	HRFT1	3460795	Pigeon River below Waterville, NC near Hartford, TN	Pigeon River	Pigeon River	538.0	1997 - Present	18.88	39,000	9/17/2004	27.03	N/A	9/27/2024	Yes
3	HEPN7	3459500	Pigeon River near Hepco, NC	Pigeon River	Pigeon River	350.0	1876 - Present	18.00	42,000	06/1876	21.59	55,200	9/27/2024 16:30	Yes
4	CTPN7	3456991	Pigeon River near Canton, NC	Pigeon River	Pigeon River	130.0	1985 - Present	22.8	48,000	9/17/2004	26.10*	N/A	9/27/2024	Yes
5	CWPN7	3456100	West Fork Pigeon River at Bethel, NC	West Fork Pigeon River	Pigeon River	58.4	1981 - Present	16.27	19,000	9/17/2004	16.78	21,200	9/27/2024 10:25	Yes
6	CEPN7	3456500	East Fork Pigeon River near Canton, NC	East Fork Pigeon River	Pigeon River	51.5	1955 - Present	16.03	19,800	8/17/2021	14.10	15,900	9/27/2024 10:25	No

*Estimated stage using HWM elevation near gage, The gage malfunctioned during Hurricane Helene

HURRICANE HELENE – POST FLOOD REPORT

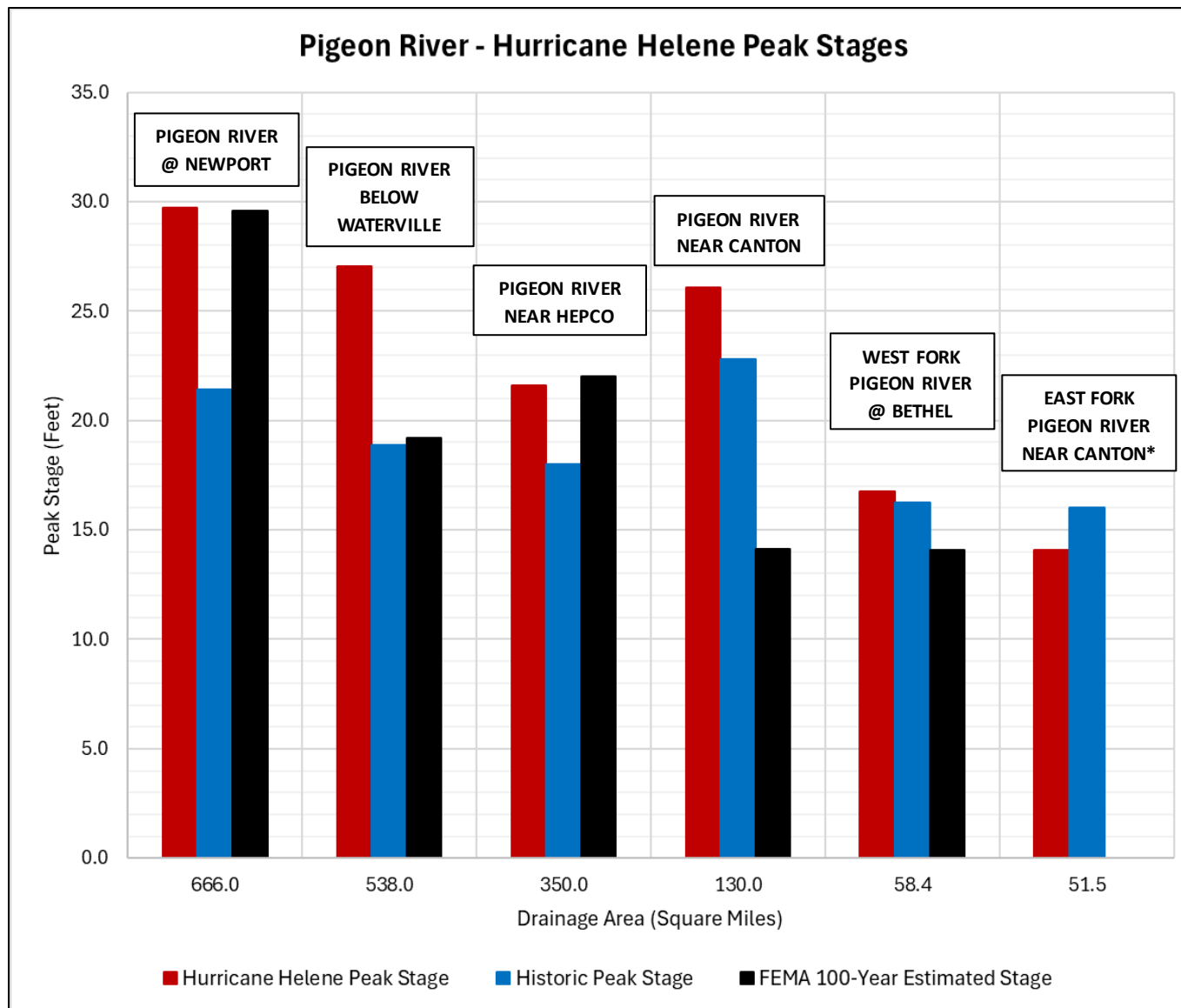


Figure 3.18: Pigeon River Watershed - Peak Stages

*This gage location does not have a FEMA effective study to reference for a 100-Year stage estimation

HURRICANE HELENE – POST FLOOD REPORT

3.4.2 French Broad River Watershed

Hurricane Helene brought record-breaking flooding to the French Broad River watershed, similar to the historic flooding experienced by the Pigeon River. The storm's impact was evident at several key locations, where flood levels exceeded previous records. The USGS Gage 03447687 French Broad River near Fletcher, NC, recorded a peak stage of 30.46 feet (shown in Figure 3.19), exceeding the previous record set in 2004 by 10.34 feet. In Asheville, NC, the French Broad River (USGS Gage 03451500) peak stage reached 24.67 feet (displayed in Figure 3.20), surpassing the 1916 flood record by 1.57 feet. Similarly, the Swannanoa River at Biltmore, NC, broke its 1916 record by 6.98 feet.

Further downstream, at USGS Gage 03455000 French Broad River near Newport, TN (Figure 3.21), the flood stage peaked at 23.34 feet, just shy of the historic 1867 flood peak of 24 feet. The data from these locations, summarized in Table 3.4 and Figure 3.22, reveals that five of the eight French Broad River gages recorded higher peak stages during Hurricane Helene than any previous flood on record. These record flood levels demonstrate the significance of the storm on the French Broad Watershed.

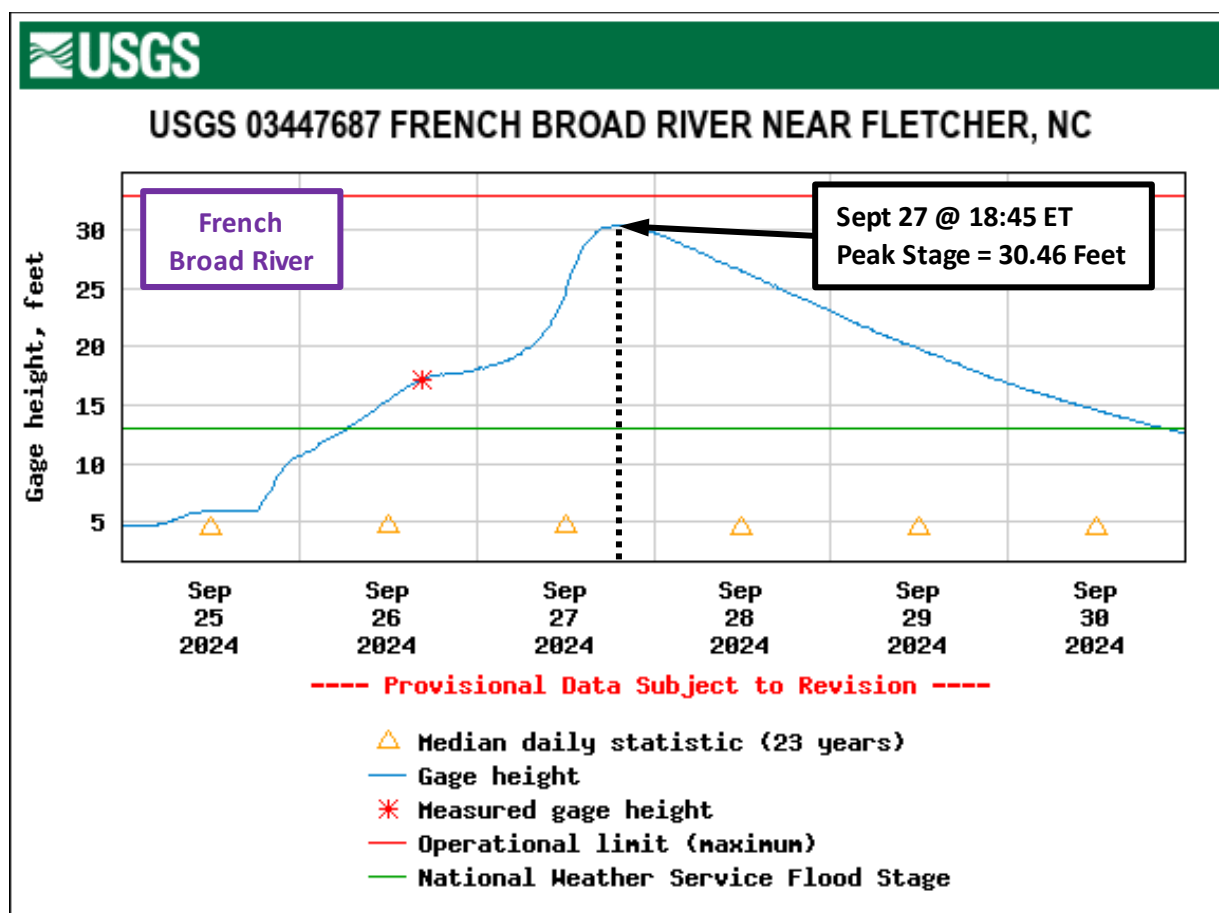


Figure 3.19: USGS Gage 03447687 French Broad River near Fletcher, NC – Stage Hydrograph

HURRICANE HELENE – POST FLOOD REPORT

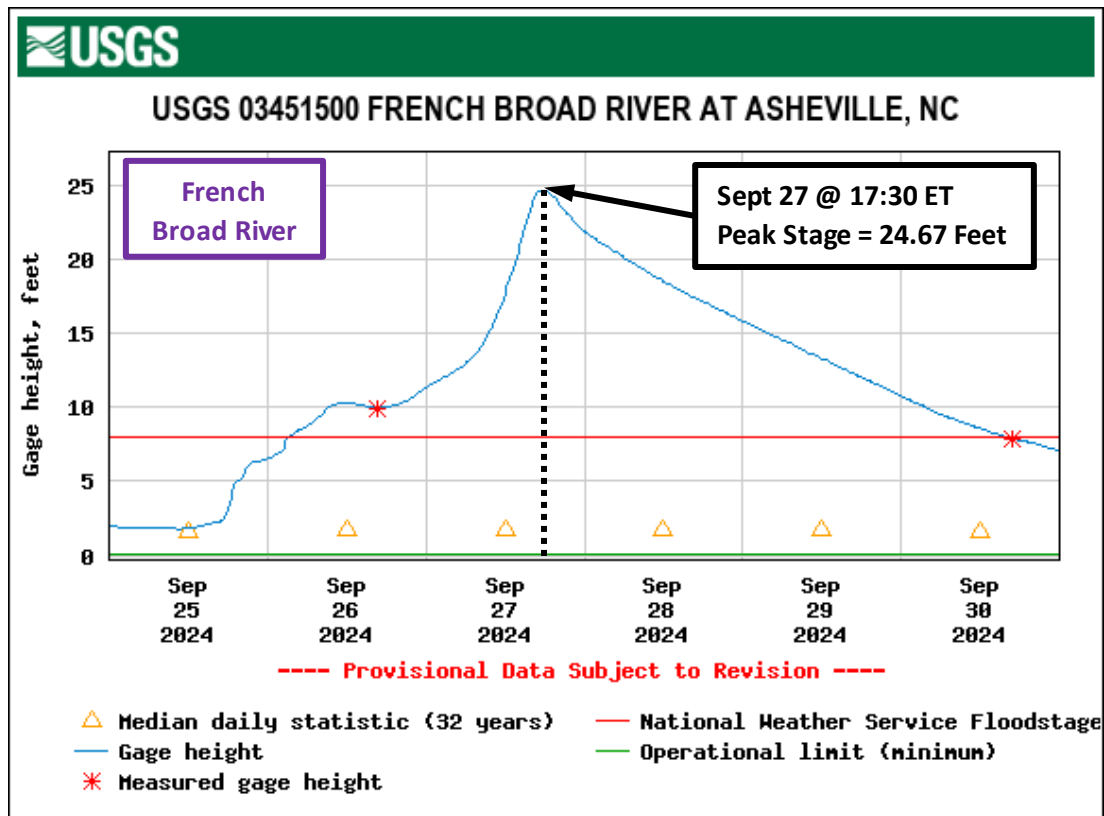


Figure 3.20: USGS Gage 03451500 French Broad River at Asheville, NC – Stage Hydrograph

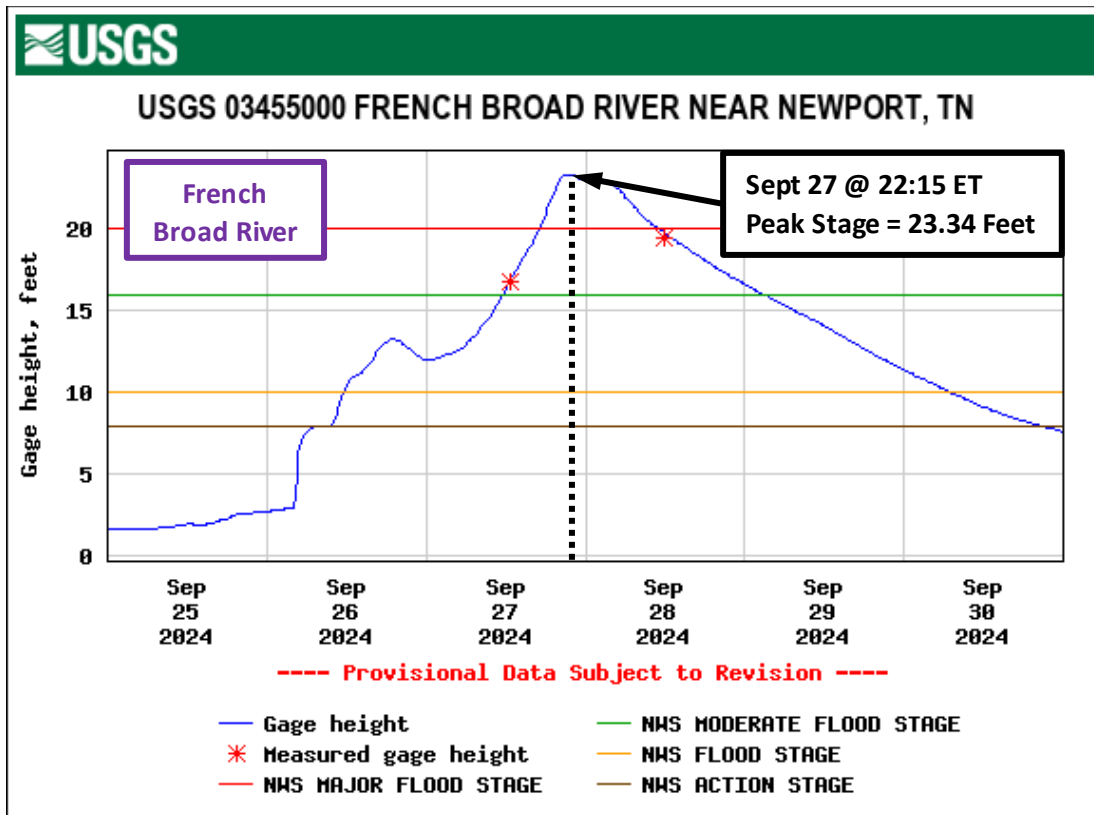


Figure 3.21: USGS Gage 03455000 French Broad River near Newport, TN – Stage Hydrograph

HURRICANE HELENE – POST FLOOD REPORT

Table 3.4: French Broad River Watershed - Gage Data

FRENCH BROAD RIVER WATERSHED – STREAMFLOW GAGE TABLE														
Gage Map Number	NWS Gage ID	USGS Gage ID	Gage Name	Stream	Watershed	Drainage Area (sq. miles)	Period of Record	Historic Peak Stage and Discharge before Hurricane Helene Flood			Hurricane Helene Peak Stage and Discharge			Helene New Historic Peak?
								Peak Stage (Feet)	Peak Discharge (CFS)	Date of Peak	Peak Stage (Feet)	Peak Discharge (CFS)	Date and Time of Peaks (ET)	
7	NWPT1	3455000	French Broad River near Newport, TN	French Broad River	French Broad River	1,858.0	1867 - Present	24.00	110,000	03/07/1867	23.34	N/A	9/27/24 22:15	No
8	HSPN7	3454500	French Broad River at Hot Springs, NC	French Broad River	French Broad River	1,567.0	1796 - Present	22.00	110,000	7/16/1916	22.27	133,000	9/27/24 15:45	Yes
9	MARN7	3453500	French Broad River at Marshall, NC	French Broad River	French Broad River	1,332.0	1916 - Present	22.00	115,000	7/16/1916	21.60*	N/A	9/27/24 20:00	No
10	AVLN7	3451500	French Broad River at Asheville, NC	French Broad River	French Broad River	945.0	1896 - Present	23.10	110,000	7/16/1916	24.67	N/A	9/27/24 17:30	Yes
11	FLCN7	3447687	French Broad River near Fletcher, NC	French Broad River	French Broad River	640.0	2001 - Present	20.12	25,500	9/8/2004	30.46	N/A	9/27/24 18:45	Yes
12	BLAN7	3443000	French Broad River at Blantyre, NC	French Broad River	French Broad River	296.0	1875 - Present	27.10	N/A	7/16/1916	28.43	N/A	9/27/2024	Yes
13	ROSN7	3439000	French Broad River at Rosman, NC	French Broad River	French Broad River	67.9	1908 - Present	14.95	13,500	10/4/1964	14.74	12,800	9/27/24 11:30	No
14	BLTN7	3451000	Swannanoa River at Biltmore, NC	Swannanoa River	French Broad River	130.0	1876 - Present	20.70	23,000	7/16/1916	27.68*	N/A	9/27/2024	Yes

*Estimated stage using HWM elevation near gage - The gage malfunctioned during Hurricane Helene

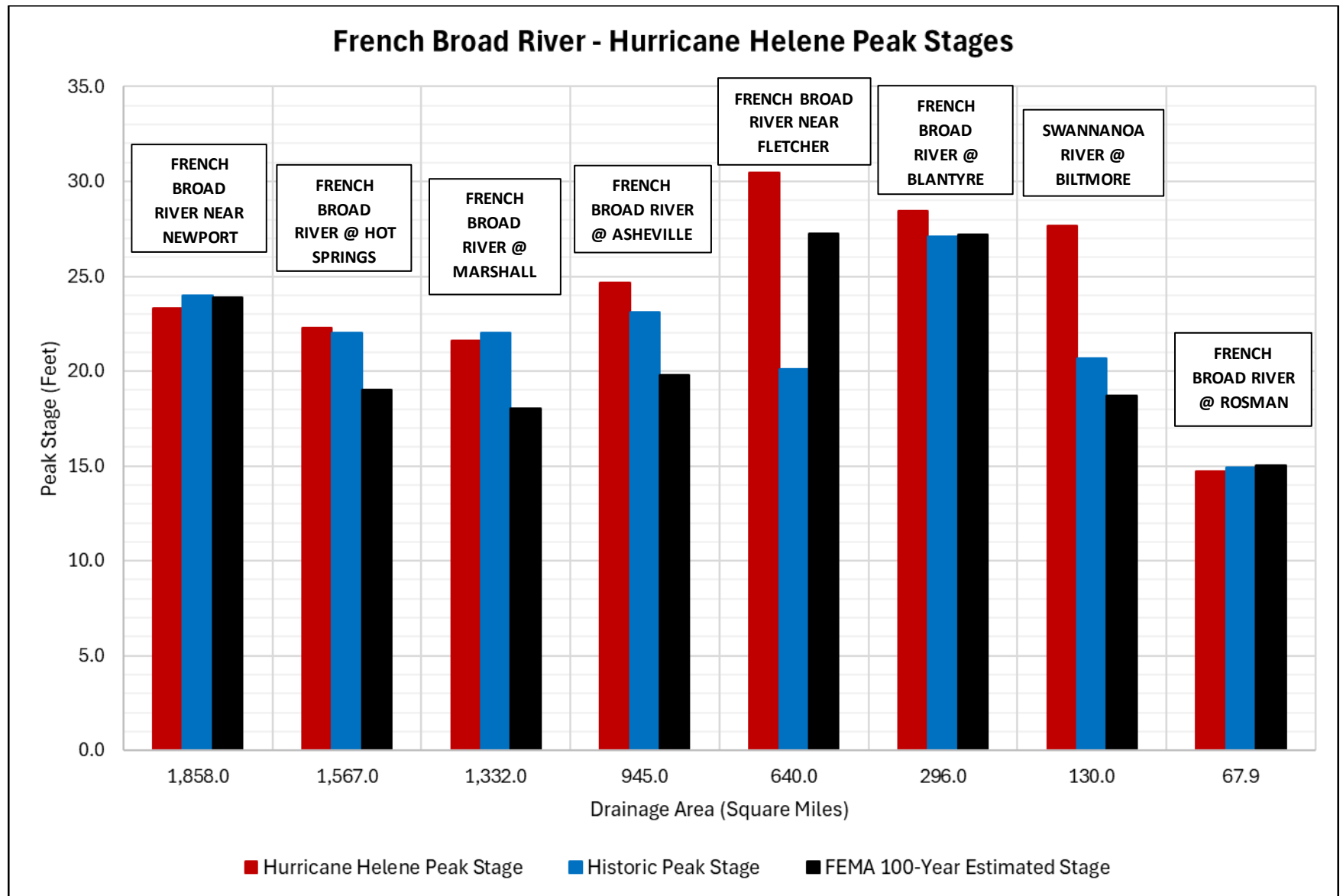


Figure 3.22: French Broad River - Peak Stages

HURRICANE HELENE – POST FLOOD REPORT

3.4.3 Nolichucky River Watershed

The Nolichucky River watershed also saw record flood levels across many of its streams in North Carolina and Tennessee. All five USGS gages in the watershed recorded peak stage for Hurricane Helene that exceeded the gages' previous record. At the Nolichucky River at Embreeville, TN gage (USGS 03465500), the peak stage was estimated to be 33.49 feet, which was 9.88 feet higher than the record held by the May 1901 flood at 24.0 feet. Below the Nolichucky River dam near Lowland, TN (USGS 03467609), the peak stage reached 35.16 feet (shown in Figure 3.23), well above (8.98 feet) the record set in February 2020 at 26.18 feet. Historic flooding not only occurred on the Nolichucky River but also many of its tributaries including Big Limestone Creek and the South Toe River. For a detailed summary of the gage data, including historic peaks and FEMA 100-year estimated stages, refer to Table 3.5 and Figure 3.24, which provide a comprehensive overview of the flooding in the Nolichucky River watershed during Hurricane Helene.

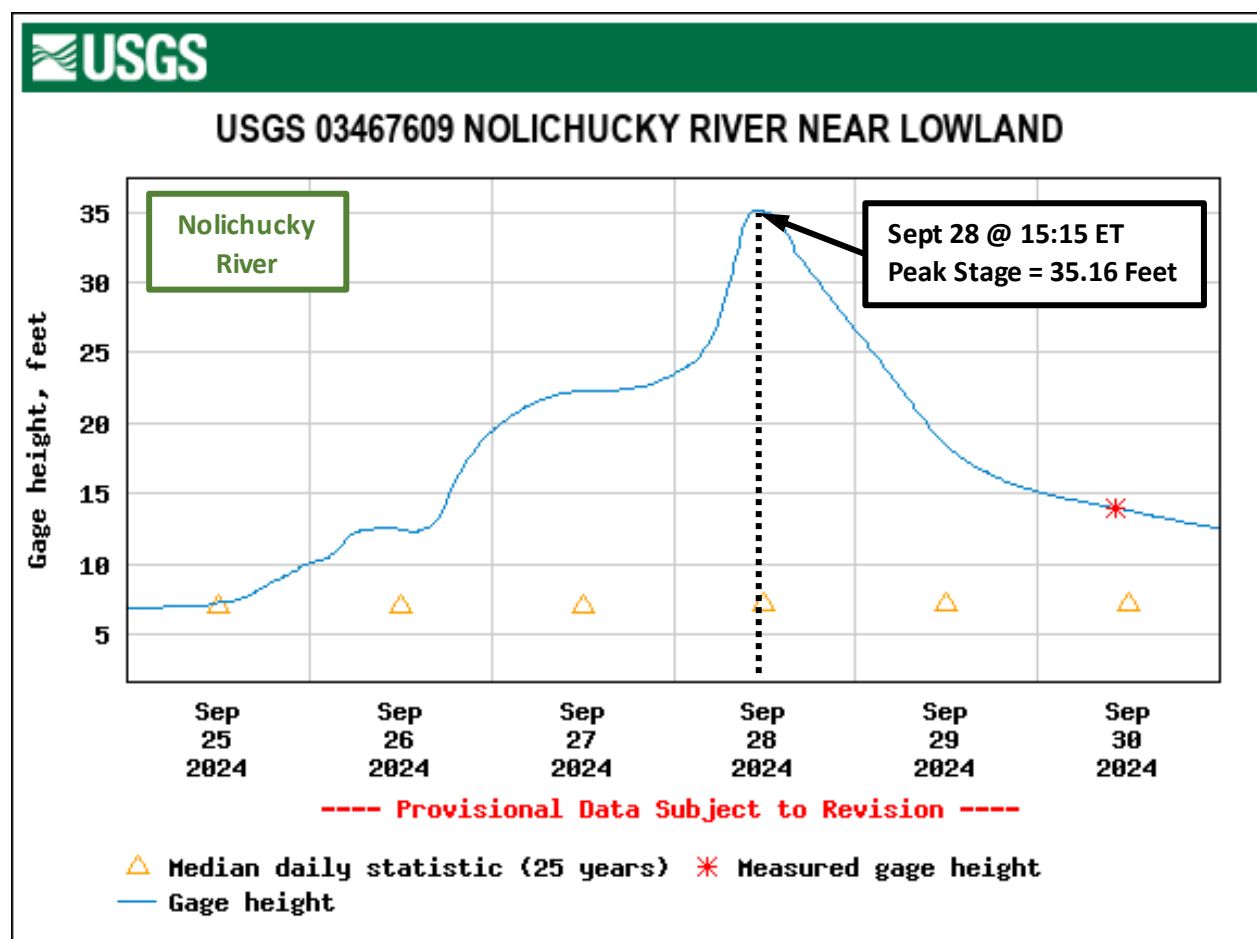


Figure 3.23: USGS Gage 03467609 Nolichucky River near Lowland, TN – Stage Graph

HURRICANE HELENE – POST FLOOD REPORT

Table 3.5: Nolichucky River Watershed - Gage Data

NOLICHUCKY RIVER WATERSHED – STREAMFLOW GAGE TABLE														
Gage Map Number	NWS Gage ID	USGS Gage ID	Gage Name	Stream	Watershed	Drainage Area (sq. miles)	Period of Record	Historic Peak Stage and Discharge before Hurricane Helene Flood			Hurricane Helene Peak Stage and Discharge			Helene New Historic Peak?
								Peak Stage (Feet)	Peak Discharge (CFS)	Date of Peak	Peak Stage (Feet)	Peak Discharge (CFS)	Date and Time of Peaks (ET)	
15	LLDT1	3467609	Nolichucky River near Lowland, TN	Nolichucky River	Nolichucky River	1,688.4	2002 - Present	26.18	48,100	2/7/2020	35.16	N/A	9/28/24 15:15	Yes
16	NOLT1	3466500	Nolichucky River Below Nolichucky Dam, TN	Nolichucky River	Nolichucky River	1,184.0	1901 - Present	51.68	44,000	2/7/2020	66.03	N/A	9/28/2024	Yes
17	LIMT1	3466208	Big Limestone Creek near Limestone, TN	Big Limestone Creek	Nolichucky River	79.0	1996 - Present	12.33	10,400	8/4/2001	18.00*	N/A	9/27/2024	Yes
17	EMBT1	3465500	Nolichucky River at Embreeville, TN	Nolichucky River	Nolichucky River	805.0	1901 - Present	24.00	120,000	5/1/1901	33.88	N/A	9/27/2024	Yes
18	STRN7	3463300	South Toe River near Celo, NC	South Toe River	Nolichucky River	43.3	1957 - Present	17.41	32,900	11/6/1977	26.06	N/A	9/27/2024	Yes

*Estimated stage using HWM elevation near gage, The gage malfunctioned during Hurricane Helene

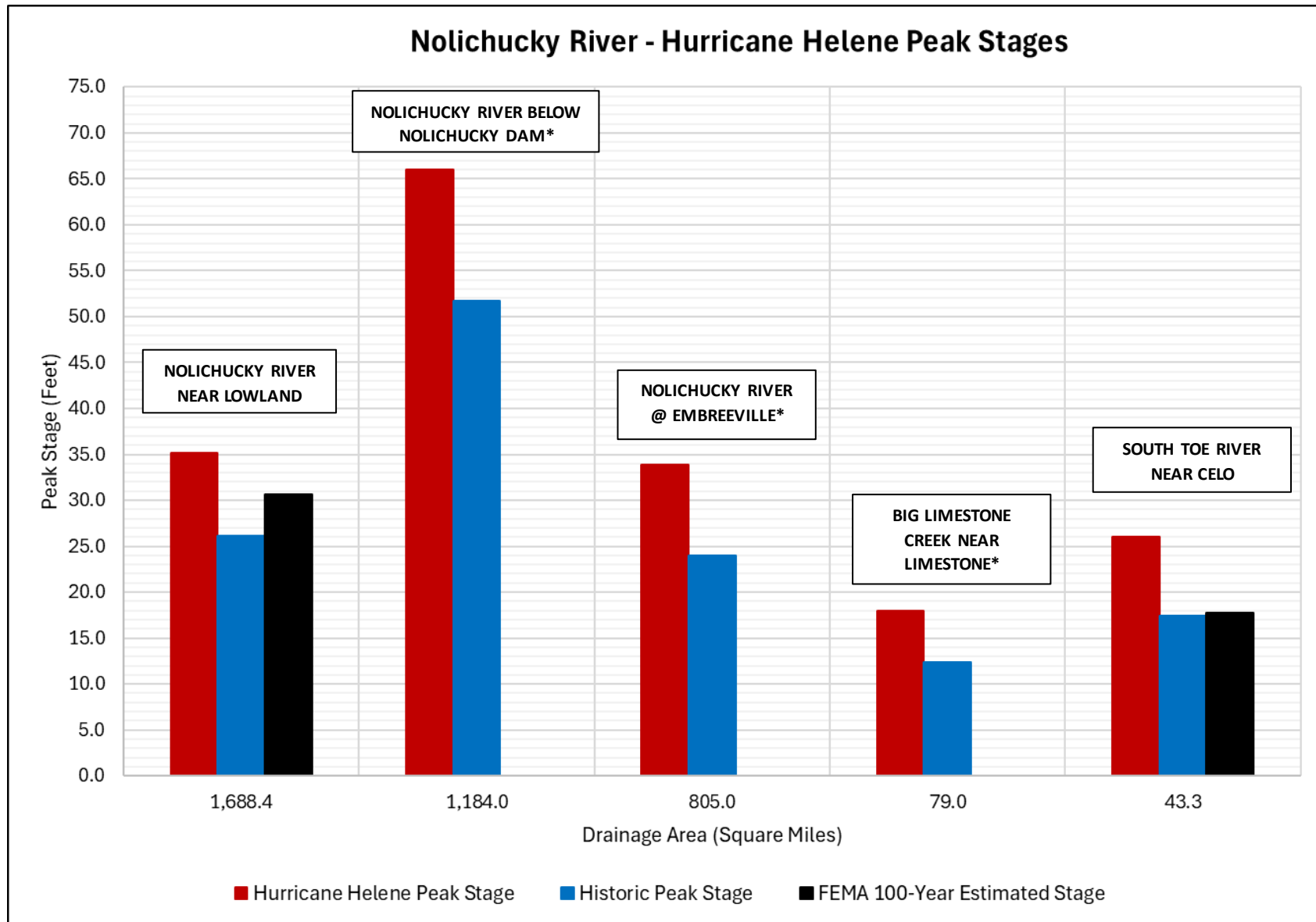


Figure 3.24: Nolichucky River - Peak Stages

*These gage locations do not have a FEMA effective study to reference for a 100-Year stage estimation

HURRICANE HELENE – POST FLOOD REPORT

3.4.4 Watauga, South Fork Holston, Tuckasegee, and Little Tennessee River Watersheds

Catastrophic flooding also occurred on several other major watersheds in North Carolina, Tennessee, and Virginia, including the Watauga, South Fork Holston, and Tuckasegee River watersheds. Notably, many of the stream gages listed in Table 3.6 and illustrated in Figure 3.29 reached or nearly reached their historic peak stages, highlighting the severity of the flooding in these areas and the widespread impacts of Hurricane Helene.

During the flood event, all three gages in the Watauga River watershed exceeded the estimated FEMA 100-Year flood stage, with two gages (USGS Gage 03486000 Watauga River at Elizabethton, TN and USGS Gage 03485500 Doe River at Elizabethton, TN) surpassing their previous historic record stages. As illustrated in Figure 3.25, the Watauga River at Elizabethton, TN stage hydrograph for Hurricane Helene shows a significant peak stage of 13.57 feet, breaking the historic peak stage of 10.70 feet by 2.87 feet set by the historic flood of May 1901. This gage is located downstream of the Watauga Lake Dam, which has been operational since 1949 and may have influenced the recorded peak stage during Hurricane Helene.

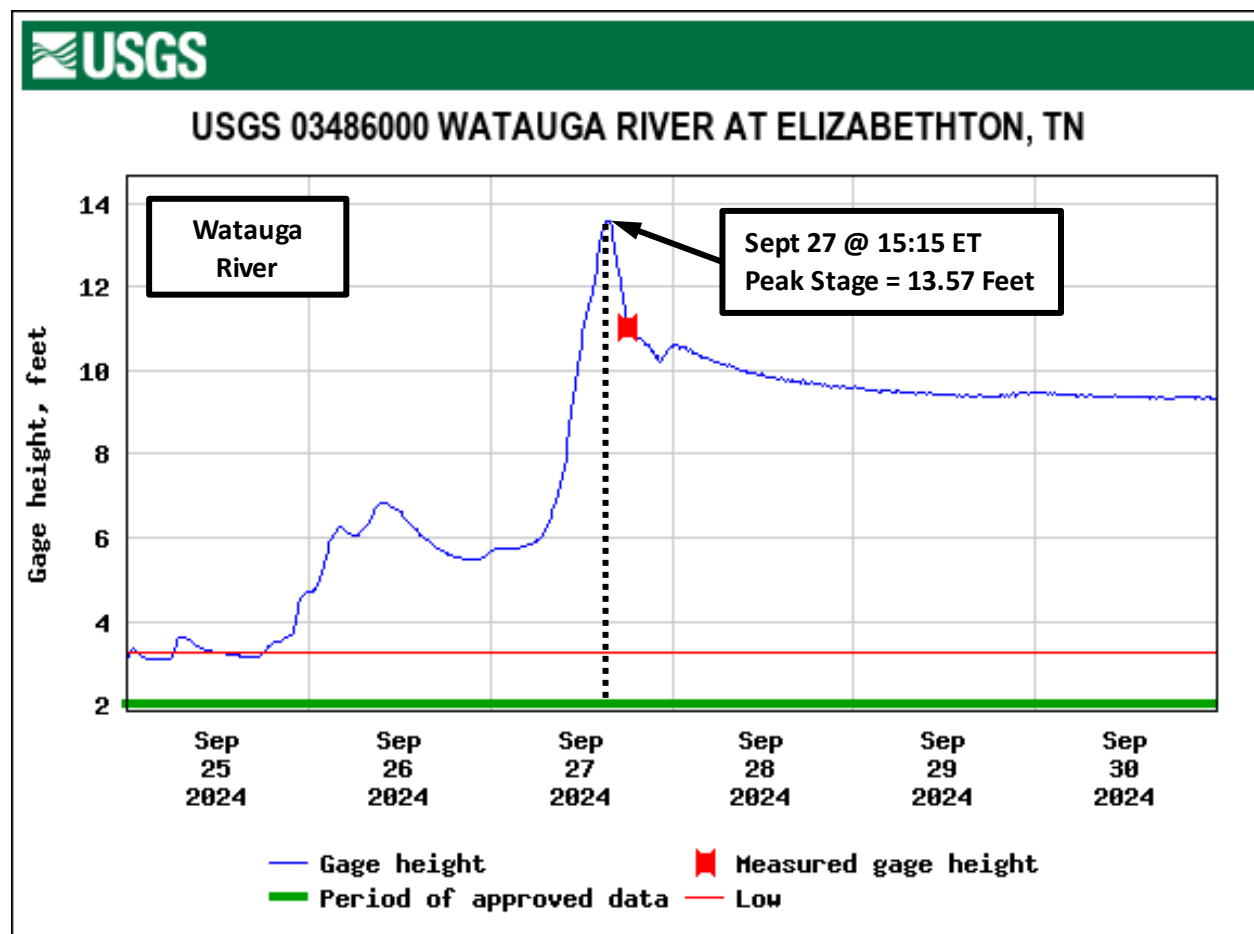


Figure 3.25: USGS Gage 03486000 Watauga River at Elizabethton, TN – Stage Hydrograph

HURRICANE HELENE – POST FLOOD REPORT

Severe historic flooding also occurred in the South Fork Holston River watershed in southwest Virginia and northeast Tennessee. Notably, Hurricane Helene set a new record stage at the Laurel Creek 1SE Damascus, VA National Weather Service (NWS) gage (DMCV2), reaching 18.0 feet and nearly doubling the previous record of 9.90 feet set in November 2003. Additionally, the storm brought the South Fork Holston River near Damascus, VA (USGS Gage 03473000) to a peak stage of 18.73 feet, as shown in Figure 3.26, falling just 1.27 feet short of the all-time record of 20.0 feet set during the historic 1867 flood.

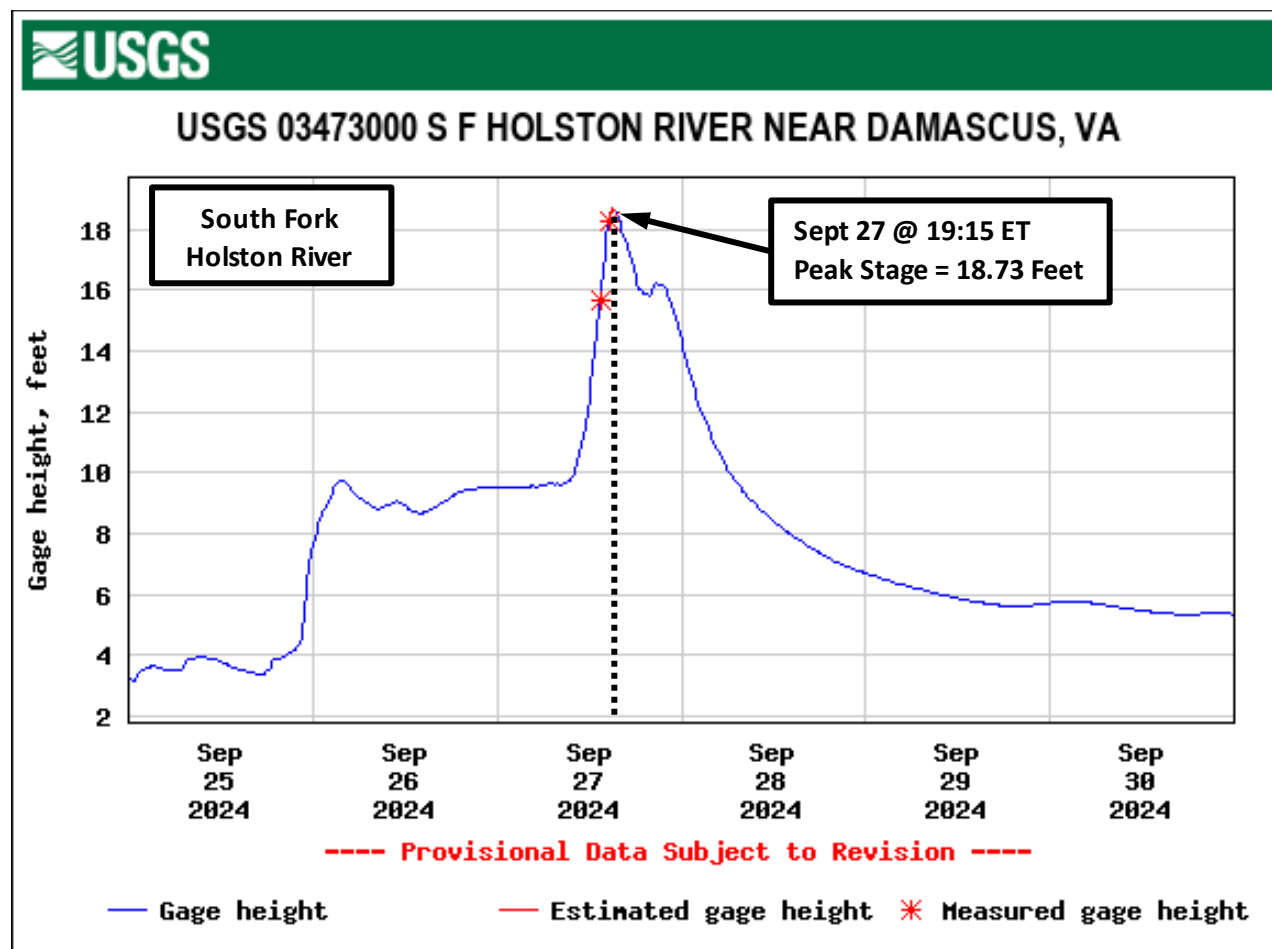


Figure 3.26: USGS Gage 03473000 South Fork Holston River near Damascus, VA – Stage Graph

The Tuckasegee River in Jackson and Swain County, NC received major flooding due to Hurricane Helene. One of the four USGS gages in the Tuckasegee River watershed recorded a new peak stage due to this flood. This gage was the Tuckasegee River at Barker's Creek, NC (USGS Gage 03510577), where the river reached a peak stage of 14.11 feet (displayed in Figure 3.27) overtaking the previous historic stage 13.26 feet set in September 2004 by 0.85 feet. Further downstream at USGS Gage 03513000 Tuckasegee River at Bryson City, NC in Figure 3.28 the flood stage peaked at 15.16 feet, which was 6.84 feet lower than the historic 1840 flood at stage 22.00 feet.

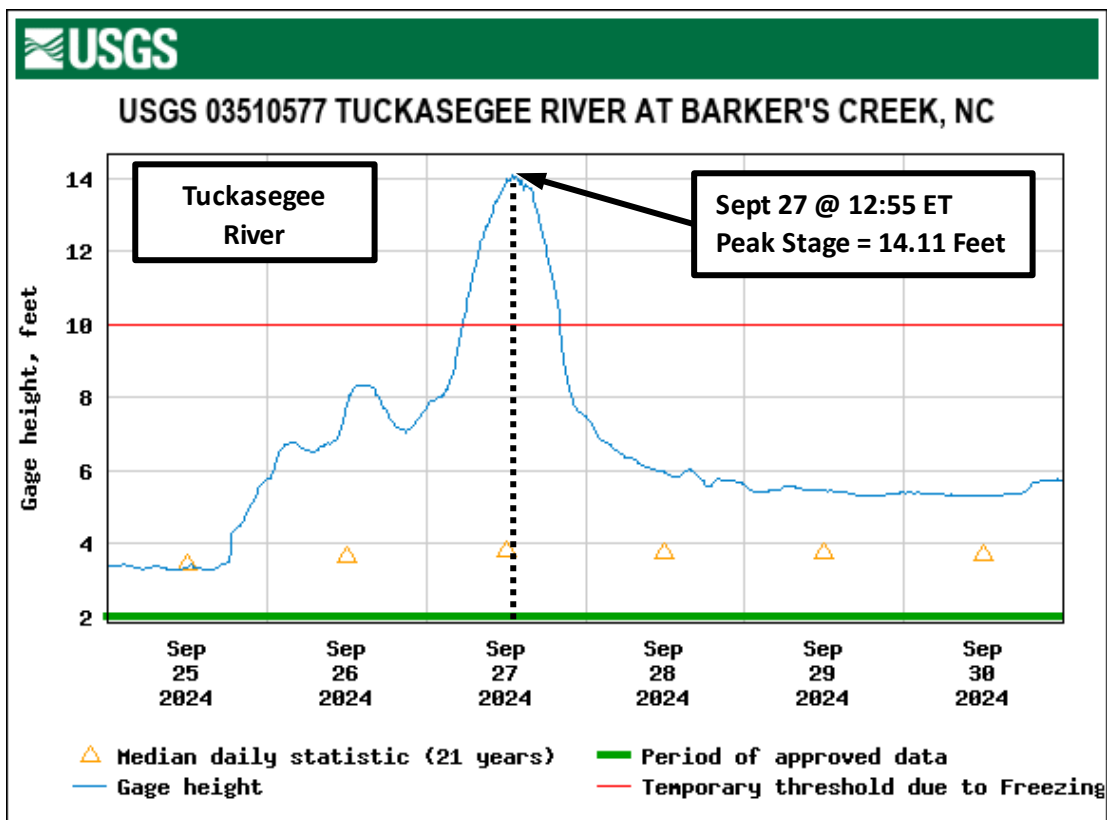


Figure 3.27: USGS Gage 03510577 Tuckasegee River at Barker's Creek, NC – Stage Graph

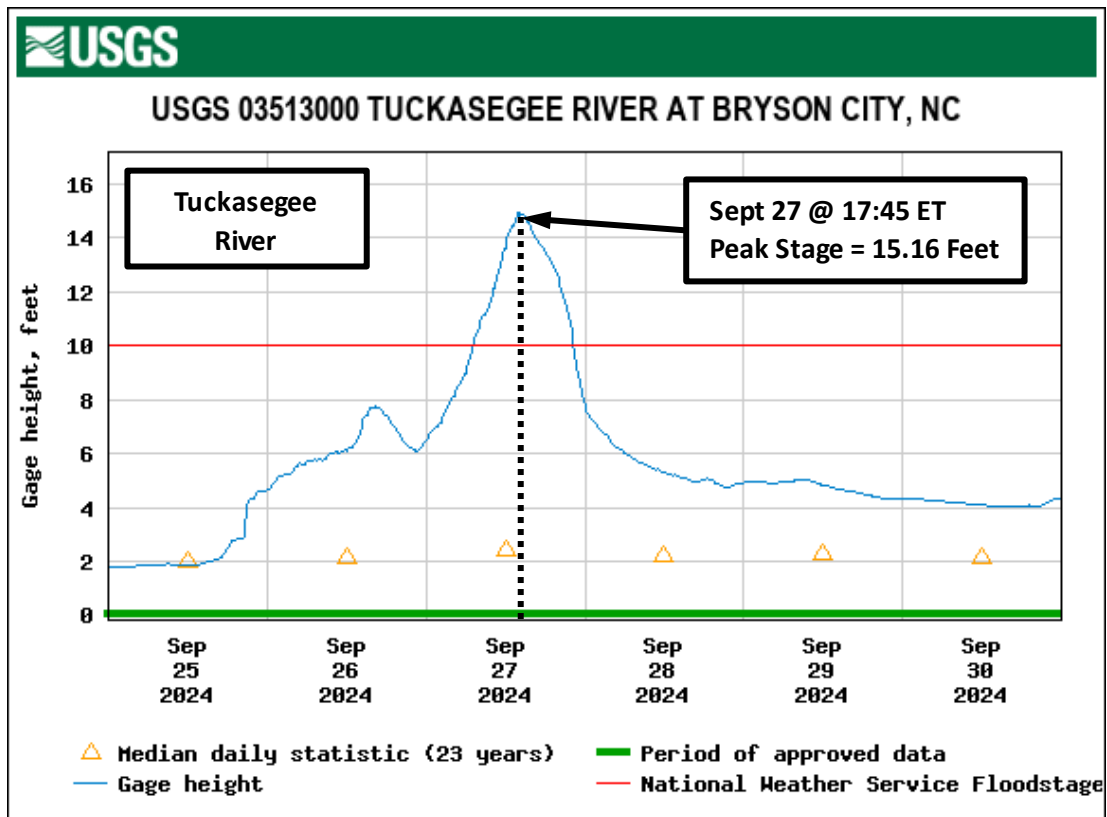


Figure 3.28: USGS Gage 03513000 Tuckasegee River at Bryson City, NC – Stage Graph

HURRICANE HELENE – POST FLOOD REPORT

Table 3.6: Watauga, South Fork Holston, and Tuckasegee River Watersheds - Gage Data

WATAUGA, SOUTH FORK HOLSTON, AND TUCKASEGEE RIVER WATERSHEDS – STREAMFLOW GAGE TABLE														
Gage Map Number	NWS Gage ID	USGS Gage ID	Gage Name	Stream	Watershed	Drainage Area (sq. miles)	Period of Record	Historic Peak Stage and Discharge before Hurricane Helene Flood			Hurricane Helene Peak Stage and Discharge			Helene New Historic Peak?
								Peak Stage (Feet)	Peak Discharge (CFS)	Date of Peak	Peak Stage (Feet)	Peak Discharge (CFS)	Date and Time of Peaks (ET)	
19	EZBT1	3486000	Watauga River at Elizabethton, TN	Watauga River	Watauga River	692.0	1867 - Present	10.7 ¹	14,500 ¹	5/21/1901	13.57 ¹	26,600 ¹	9/27/24 15:15	Yes
20	DOET1	3485500	Doe River at Elizabethton, TN	Doe River	Watauga River	137.0	1901 - Present	10.92	N/A	1/8/1998	11.10	N/A	9/27/24 18:30	Yes
21	SGWN7	3479000	Watauga River near Sugar Grove, NC	Watauga River	Watauga River	92.1	1916 - Present	29.60	50,800	8/13/1940	29.32*	N/A	9/27/2024	No
22	VSTV2	3473000	South Fork Holston River near Damascus, VA	South Fork Holston River	South Fork Holston River	303.0	1867 - Present	20.00	26,500	03-1867	18.73	26,100	9/27/24 19:15	No
23	DMCV2	N/A	Laurel Creek 1SE Damascus, VA	Laurel Creek	South Fork Holston River	97.4	2003 - Present	9.90	N/A	11/19/2003	18	N/A	9/27/2024	Yes
24	TKSN7	3513000	Tuckasegee River at Bryson City, NC	Tuckasegee River	Tuckasegee River	655.0	1840 - Present	22.00	N/A	05-1840	15.16	N/A	9/27/2024 17:45	No
25	BIRN7	3512000	Oconaluftee River at Birdtown, NC	Oconaluftee River	Tuckasegee River	184.0	1907 - Present	18.00	N/A	11/19/1906	10.18	N/A	9/27/24 12:00	No
26	DILN7	3510577	Tuckasegee River at Barker's Creek, NC	Tuckasegee River	Tuckasegee River	360.0	2004 - Present	13.26	15,800	9/17/2004	14.11	N/A	9/27/24 12:55	Yes
27	TKRN7	3508050	Tuckasegee River at SR 1172 near Cullowhee, NC	Tuckasegee River	Tuckasegee River	147.0	2004 - Present	17.37	11,500	9/8/2004	17.26	N/A	9/27/24 11:00	No

*Estimated stage using HWM elevation near gage, The gage malfunctioned during Hurricane Helene

¹ Gage Peak stage and flow affected by upstream regulation

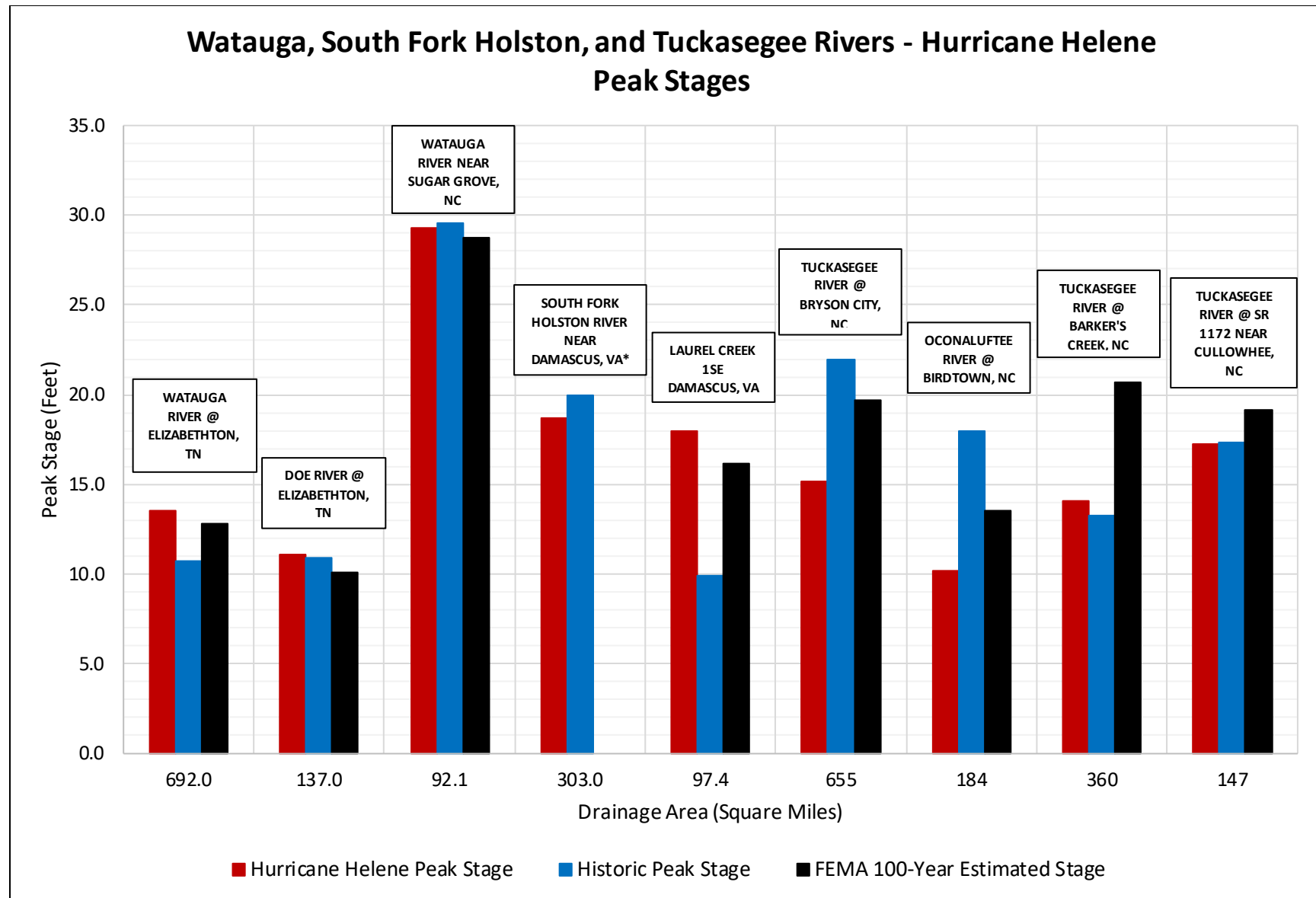


Figure 3.29: Watauga, South Fork Holston, and Tuckasegee River - Peak Stage
 *This gage location does not have a FEMA effective study to reference for a 100-Year stage estimation

4 Field Data Collection

4.1 High-Water Mark Data

As part of the post flood investigation, elevation surveys of 2,587 HWMs across 19 counties in East Tennessee, West North Carolina, and southwest Virginia, on roughly 1,300 river miles and 115 streams including the French Broad River, Pigeon River, Nolichucky River, and Watauga River were conducted by the USACE. HWM surveys were conducted from October 09, 2024, to November 23, 2024. The geographic extent of the HWM data collection can be seen in Figure 4.1 below, and a summary of HWM totals by watershed is shown in Table 4.1. HWMs were collected in six counties in East Tennessee (Cocke, Carter, Greene, Johnson, Unicoi, and Washington County), 12 counties in North Carolina (Avery, Buncombe, Haywood, Henderson, Jackson, Macon, Madison, Mitchell, Swain, Transylvania, Watauga, and Yancey County), and 1 county in Virginia (Washington County). Of the 2,587 HWMs collected, 2,193 were collected in North Carolina, 351 in Tennessee, and 43 in Virginia.

Table 4.1: Number of HWMs Collected per Watershed

Watershed	Number of HWMs Collected
Nolichucky River	731
French Broad River	888
Watauga River	251
Pigeon River	530
Tuckasegee River	124
South Fork Holston River	56
Little Tennessee River	7

The Nashville District had previously developed a document outlining the standard operating procedure (SOP) for collecting HWMs as well as other references for preparing to collect HWMs and collecting HWMs (USACE Nashville District - Water Resources Section, 2024). These procedures and materials were used and referenced for this effort.

The subsequent sections will discuss the importance of HWM collection following an extreme flood event, the pre-deployment preparation for collection, the standard field operating procedures, and the post-processing of HWM data.

HURRICANE HELENE – POST FLOOD REPORT

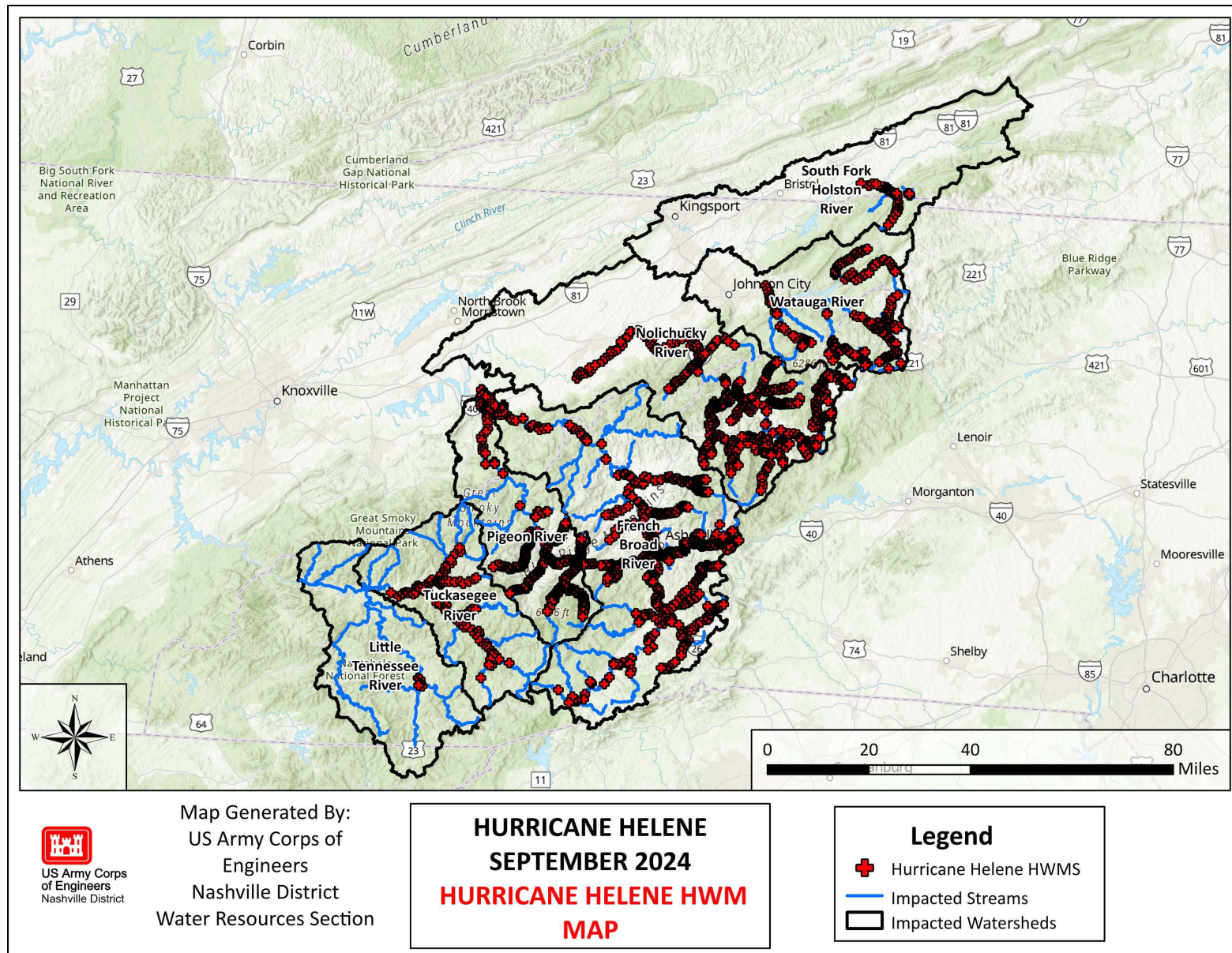


Figure 4.1: Hurricane Helene – High-Water Mark (HWM) Overview Map

HURRICANE HELENE – POST FLOOD REPORT

4.1.1 High-Water Mark Significance

HWMs are reference points set to document the maximum height at which water rose along a waterway during a flood event. They are based on physical evidence left behind by the flood and should be collected at the earliest safe opportunity after floodwaters have receded. Examples of HWMs can include mud/seed lines on structures, debris lines in fields, and debris snags in trees or fences.

Collecting HWMs is an important part of hydrologic and hydraulic studies. Promptly collecting HWMs after an event occurs is crucial, as clean-up or other precipitation events can inadvertently remove HWMs. Further, HWMs provide key data when calibrating future models. They allow for the ability to validate the accuracy of hydrologic and hydraulic models which further provides more accurate results and mapping for the impacted areas. Figure 4.2 to Figure 4.5 below provide some examples of HWMs the team collected throughout North Carolina and Tennessee.



Figure 4.2: Mud Line on Building along the Swannanoa River, NC



Figure 4.3: Debris Snag in Fence along the Doe River, TN

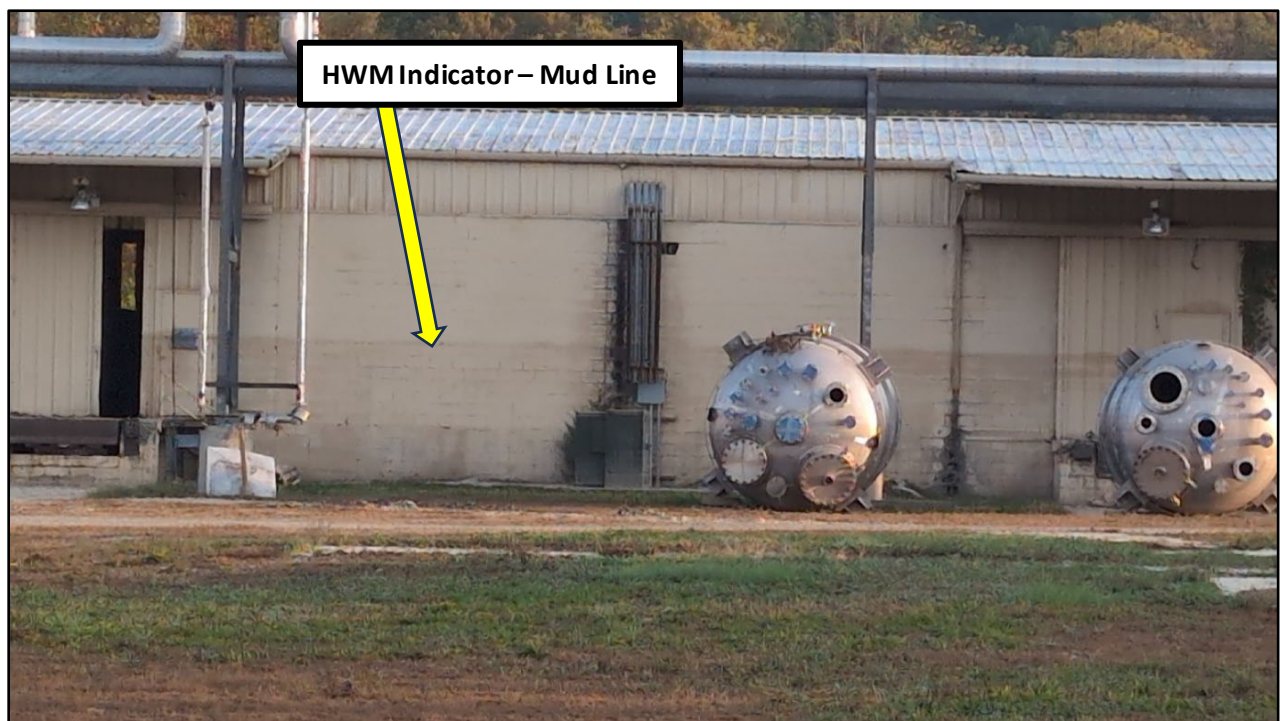


Figure 4.4: Mud Line on Building along the Pigeon River, TN

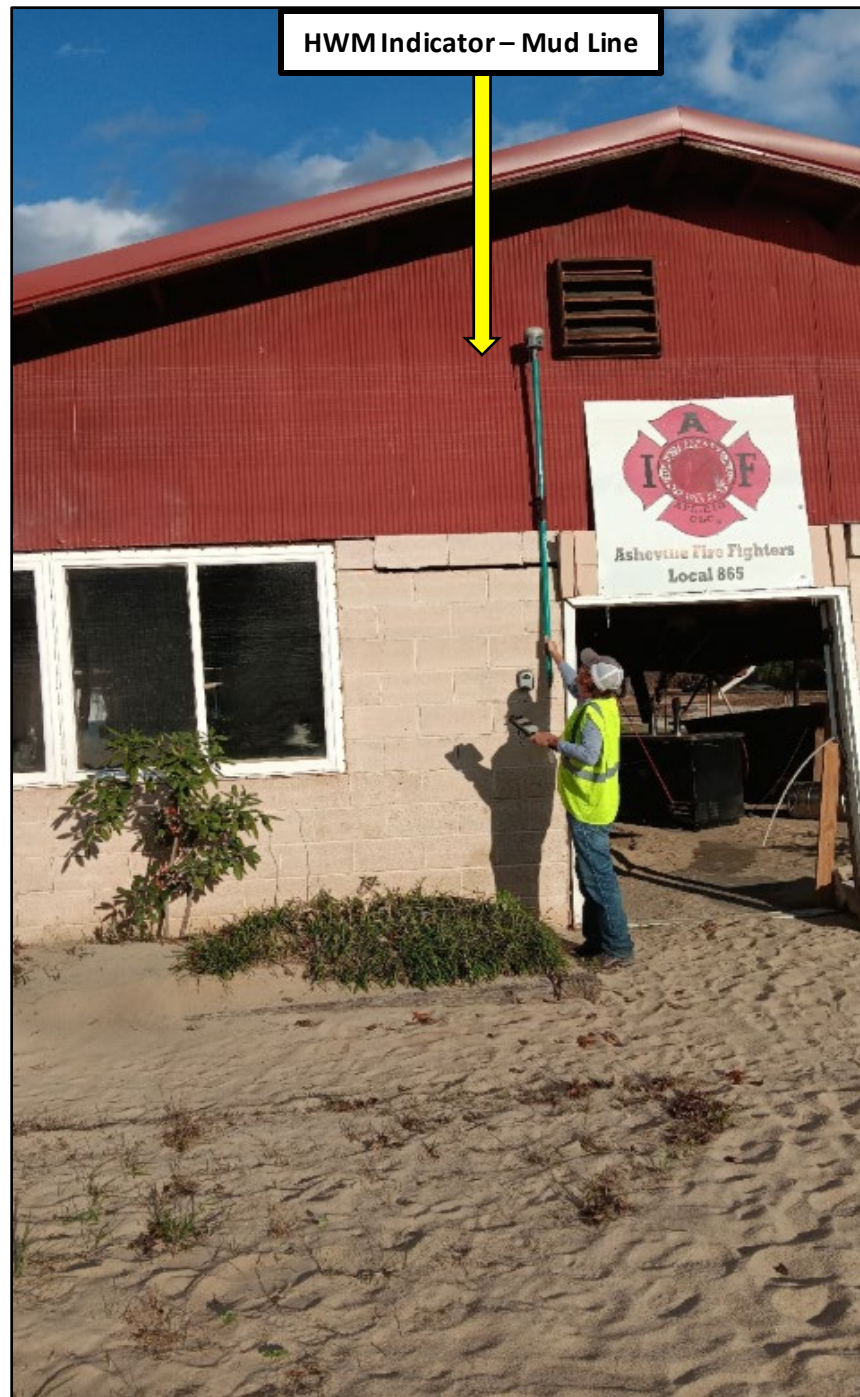


Figure 4.5: Mud Line on Building along the French Broad River, NC

4.1.2 High-Water Mark Pre-Deployment Preparation

Pre-deployment planning and activities involve carefully determining where to collect post-flood data, identifying the right personnel to carry out the collection, and establishing clear methods and tools for efficient and accurate data gathering in the field.

Streams and areas of interest (AOIs) for HWM data collection had to be determined before teams were deployed. Streams of interest were determined by selecting areas with known significant damage. Some

HURRICANE HELENE – POST FLOOD REPORT

of these areas were able to be validated by the NOAA aerial imagery that was flown immediately after the event, as seen in Figure 4.6 below of the before and after imagery in Asheville, NC (National Oceanic and Atmospheric Administration, 2024).

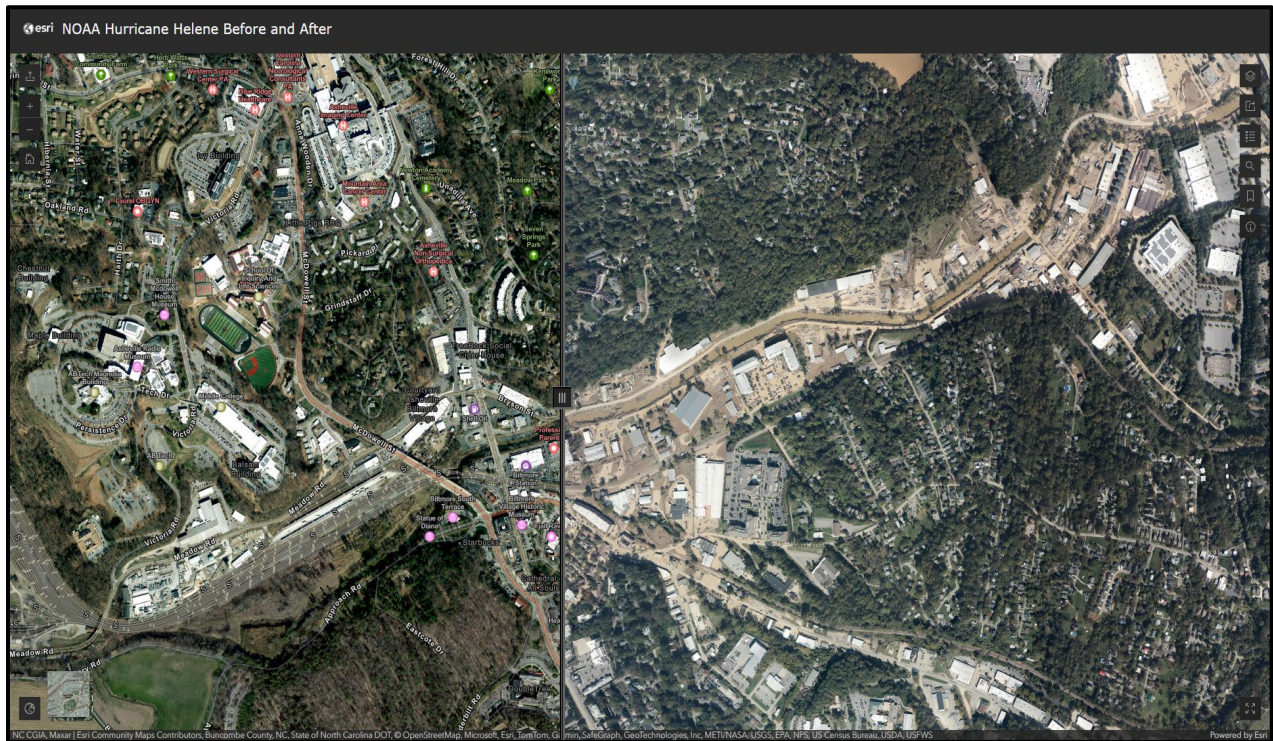


Figure 4.6: NOAA Aerial Imagery Before and After - Asheville, NC (*National Oceanic and Atmospheric Administration, 2024*)

In areas where imagery was unavailable, streams and areas of interest were identified based on where HWMs may easily be found (e.g., mud/seed lines on buildings or debris snags on fences). These areas of interest were documented in ArcGIS, reviewed by the HWM team lead, and final points were uploaded in ArcGIS Field Maps which could be accessed by all team members via the ArcGIS Field Maps application. Field Maps is a tool that allowed for real-time collaboration between the office support team and field teams. Areas of interest that were uploaded provided field teams with coordinates to where HWMs were likely to be found. Similarly, field teams could upload the general location of the points they collected, with pictures attached, so the office support team could easily track the progress being made. Other areas and streams of interest were occasionally identified by field teams with first-hand accounts of the flooding and damage.

The formation of well-coordinated field teams, each composed of personnel with specialized expertise, is essential to ensure the successful and accurate collection of post-flood data in the field. Each HWM team consisted of at least three members: an H&H team lead, a member from the surveying section, and an additional data logger to support efficient documentation. Teams coordinated daily with each other to ensure they pursued dense yet not duplicative collection of each watershed. The responsibilities of each field team are described in the section 4.1.3.

HURRICANE HELENE – POST FLOOD REPORT

4.1.3 High-Water Mark Data Collection Process

As previously noted in section 4.1, the Nashville District recently developed a SOP and accompanying documentation for the collection of flood data. Those resources were applied to this collection effort and are documented below.

Office support team members prepared reference maps in ArcGIS Field Maps with the areas of interest along streams to be surveyed. Field teams then were responsible for locating HWMs near the AOIs, determining the type of HWM they were collecting (e.g., debris snag, mud line, seed line, etc.), and recording a detailed log of associated HWM data in both data collection sheets and ArcGIS Field Maps. Each HWM was assigned a quality rating, a visual description of the HWM, a photograph, and a maximum elevation (see section 4.2 for survey procedures). Data collection sheets, shown in Figure 4.7 and Figure 4.8 below, were provided to each team to maintain a consistent data collection format.

 High-Water Mark Data Collection						
Location: _____				Date: _____		
Data Collectors: _____				Page: _____ of _____		
Name	Stream	Event	Details	Date Surveyed	Photo	Description
HWM Type			Quality		☐ Y	
			Height		# of Photos	
HWM Type			Quality		☐ Y	
			Height		# of Photos	
HWM Type			Quality		☐ Y	
			Height		# of Photos	
HWM Type			Quality		☐ Y	
			Height		# of Photos	
HWM Type			Quality		☐ Y	
			Height		# of Photos	

Figure 4.7: High-Water Mark Data Collection Sheet - Front

HURRICANE HELENE – POST FLOOD REPORT

 High-Water Mark Additional Notes	
Location: _____	Date: _____
Data Collectors: _____	
Include additional notes below. (example: comments from residents)	

Figure 4.8: High-Water Mark Data Collection Sheet - Back

The quality of a HWM depends on the amount of vertical uncertainty present in the HWM indicator. Vertical uncertainty is the range of peaks and valleys on the indicator line, measured in feet. In other words, the straighter the line of the HWM indicator, the higher the quality of the HWM. When a HWM is identified, it should be based on the best indicator in the area, and it should be verified by comparing it with other indicators in the area. The main HWM indicator should be described, and the quality of the mark should be assigned based on that main indicator. In accordance with the USGS *Field Manual for Identifying and Preserving High-Water Mark Data* (2017), quality ratings were given to each HWM for consistency among all collectors. Table 4.2 below provides the referenced USGS quality ratings.

Table 4.2: High-water Mark Quality Chart

HWM Quality and Shorthand	Amount of Vertical Uncertainty
Excellent (E)	Within ± 0.05 foot (± 0.6 -in)
Good (G)	Within ± 0.10 foot (± 1.2 -in)
Fair (F)	Within ± 0.20 foot (± 2.4 -in)
Poor (P)	Within ± 0.40 foot (± 4.8 -in)
Very Poor (V)	More than ± 0.40 foot ($> \pm 4.8$ -in)
High-water mark defines the minimum height of the peak, but peak may have been higher to an unknown extent.	At least this high (ALTH)

4.2 Surveying

4.2.1 Survey Equipment and Accuracy

HWM elevations were collected using Trimble GPS survey units by professional surveyors with the team. Trimble R12i Global Navigation Satellite Systems (GNSS) were used as the primary surveying units, supplemented by a Trimble R2 GNSS system in North Carolina.

The Trimble R12i units were equipped with Trimble GEDU Inertial Measurement Units (IMU), which are highly accurate sensors with inertial measurement technology, optimized for use with Trimble GEDO track measurement systems. These sensors allow the R12i units to be non-level and tilted (example displayed in Figure 4.9), enabling the HWMs to be collected in challenging areas like under buildings or tree cover.

The team surveyors used the Real-Time Kinematic (RTK) survey technique by connecting to the Tennessee and North Carolina State Continuously Operating Reference Station (CORS) networks. RTK uses a combination of GPS signals and additional corrections from a CORS reference station to achieve precise positioning. The CORS reference station, usually located at a known location, sends correction data to the rover (the device being used to collect data) via radio or cellular communication. The residuals of the vertical and horizontal precision were within 0.1 foot for most of the HWMs. For challenging areas and marks, the residuals were increased up to 1 foot and were noted in the additional information attached to each mark.

The accuracy and precision of the collected field survey data was verified through known reference points such as survey benchmarks or set control points (e.g. bridge monuments, external agency mark, etc.). The U.S. National Geodetic Survey (NGS) publishes a survey mark database (i.e. benchmarks) with known coordinates and elevations. This database can be used by surveyors in the field to verify that equipment is working properly and calibrated to acceptable limits. For this mission, these reference points were surveyed and verified to ensure the survey unit's collected data matched, or closely matched within reasonable bounds, the known data of the reference point. This process ensures data is collected with the utmost quality and accuracy.

While in the field, teams experienced some technical issues related to the storm damage that impacted data collection. A notable issue was the impact to communication systems (e.g. cell phone and internet service), which affected the ability to use the survey equipment and some of its services. This issue was largely temporary or remediated. Another challenge was impassable roads from mud slides, bridge removals, and debris, which limited access and produced longer travel times to some of the heavily damaged areas.



Figure 4.9: Trimble R12i Tilt of Survey Unit

4.2.2 Survey and HWM Coordinate Systems

The North American Vertical Datum of 1988 (NAVD88) was used as the vertical datum and the North American Datum (NAD) 1983 (2011) was used as the horizontal datum for all surveyed points. The Tennessee HWMs projection was NAD 1983 (2011) StatePlane Tennessee FIPS 4100 (US Feet), and the North Carolina HWMs were NAD 1983 (2011) StatePlane North Carolina FIPS 3200 (US Feet). The Virginia HWMs used the NAD 1983 (2011) StatePlane Tennessee FIPS 4100 (US Feet) due to the marks being at the border of Tennessee and Virginia. The final complete HWM database was reprojected to NAD 1983 (2011) StatePlane North Carolina FIPS 3200 (US Feet).

4.3 First-Hand Accounts of Hurricane Helene

The field team also gathered valuable first-hand accounts from residents affected by the hurricane, providing crucial insights into the extent of the damage and its impact on the community. Residents and business owners in the affected areas validated flood depths and shared personal anecdotes of their experience. Residents assisted in identifying HWMs on their properties, verified height of floodwaters, and even provided photos or videos documenting the flood.

HURRICANE HELENE – POST FLOOD REPORT

The USACE field teams offered flyers, brochures, and business cards to many constituents of the East Tennessee and western North Carolina regions to provide additional means to submit photos, videos, or other relevant information. Figure 4.10 to Figure 4.13 shown below are some of the images sent to the team by residents to aid in the current and future efforts.



Figure 4.10: Flooding in Asheville, NC



Figure 4.11: Flooding in Candler, NC

HURRICANE HELENE – POST FLOOD REPORT



Figure 4.12: Dillingham Rd - Barnardsville, NC

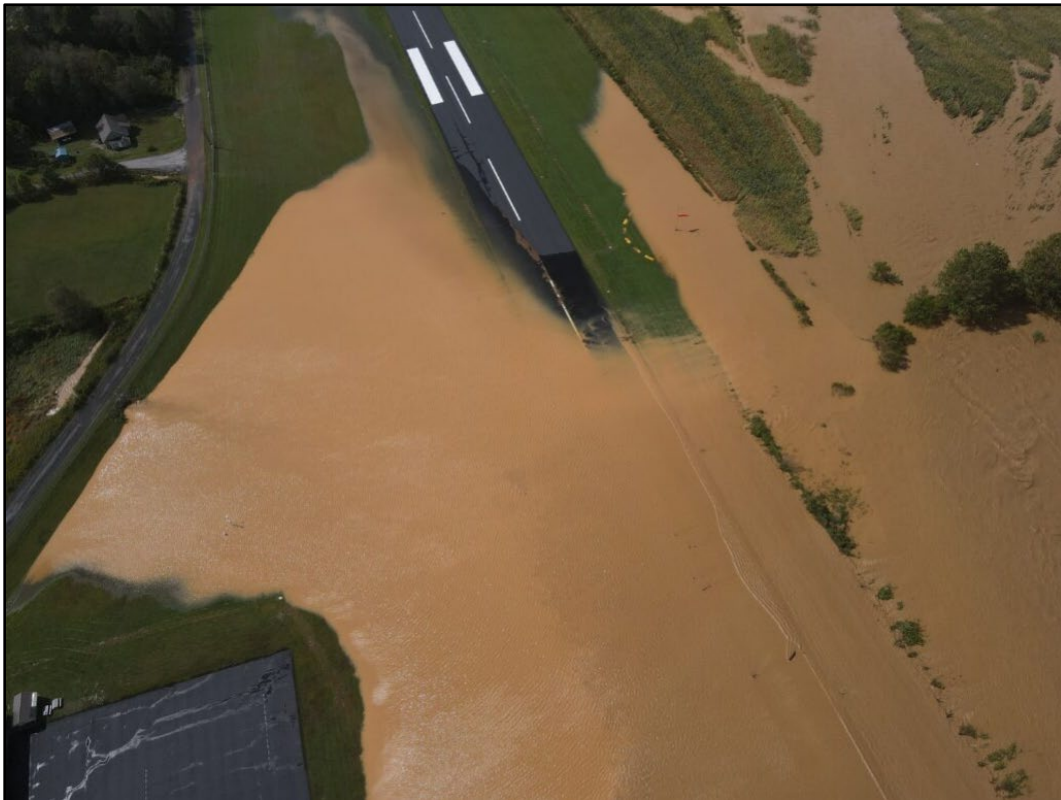


Figure 4.13: Johnson County Airport - Johnson County, TN

4.4 Post-Processing and Final Database

The HWM data was organized and compiled into a Geographical Information System (GIS) geodatabase, shapefile, Google Earth KMZ file, layer packages, and map package. The process of organizing and categorizing the data included ensuring HWM points were appropriately matched with the accompanying survey data and renamed according to position on each stream (i.e. downstream to upstream), with the surveyed points also being reprojected to the same projection. All descriptions from the field collection sheets were added and linked to the corresponding HWM. An example of the HWM data included in the collection sheets and post-processing information can be found in Appendix B: High-Water Mark Data Collection Table - Example.

5 Conclusion

Hurricane Helene was a devastating, record-breaking event for much of the western North Carolina and eastern Tennessee streams and towns in September 2024. Based on the NOAA Atlas 14 precipitation estimates, the preliminary 3-day precipitation estimates for much of western North Carolina, including Asheville, were above the 0.001 (1/1000, 1000-year) annual exceedance probability. This indicates that the 3-day rainfall totals in this region were so extreme that the statistical chances of them being exceeded in an any given year is less than 1 in 1,000. While Tennessee did not receive the same magnitude of rainfall as North Carolina, the effects were just as devastating due to many of the affected North Carolina streams flowing into eastern Tennessee watersheds. Many gages located throughout eastern Tennessee broke previous peak stage and flow records by almost 10 feet.

The severity of the event came from many factors including the steep, mountainous terrain, and the antecedent conditions from a pre-hurricane rainfall two days prior to Hurricane Helene, leaving soil conditions saturated. Furthermore, Hurricane Helene had the ideal meteorological conditions to bring heavy, tropical moisture to the region.

This document, and the data collected and developed in support of it, will provide local communities and agencies with a resource that can be utilized in future project planning, hydraulic and hydrologic analyses, and project performance data. In particular, the collection of HWMs across a large portion of western North Carolina and eastern Tennessee will provide the Corps and its partners with the information needed to perform future detailed studies on areas of interest which will ultimately help cities and agencies prepare for, and respond to, future flood risks.

6 References

- Community Collaborative Rain, Hail, and Snow Network. (2024). *CoCoRaHS*. Retrieved from <https://www.cocorahs.org/>
- National Oceanic and Atmospheric Administration. (2024). *NOAA Hurricane Helene Before and After*. Retrieved from <https://www.arcgis.com/apps/instant/media/index.html?appid=fe5193ccf48f48e3a2129b6c394ee886>
- National Oceanic and Atmospheric Administration. (n.d.). *NOAA Atlas 14 Point Precipitation Frequency Estimates*. Retrieved from https://hdsc.nws.noaa.gov/pfds/pfds_map_cont.html?bkmrk=tn
- National Oceanic and Atmospheric Administration. (2024, November). *Climate.gov*. Retrieved from <https://www.climate.gov/news-features/event-tracker/hurricane-helenes-extreme-rainfall-and-catastrophic-inland-flooding>
- National Weather Service. (2024). *National Weather Service*. Retrieved from https://www.weather.gov/mrx/Hurricane_Helene
- National Weather Service. (n.d.). *National Weather Service*. Retrieved from Flooding in North Carolina: <https://www.weather.gov/safety/flood-states-nc>
- North Carolina Geological Survey and UNC Asheville's NEMAC. (n.d.). *The Great Flood of 1916*. Retrieved from <https://storymaps.arcgis.com/stories/bcb2e5373c21491c9e68e96a8156d92b>
- U.S. Geological Survey. (2017). Field Manual for Identifying and Preserving High-Water Mark Data.
- USACE Nashville District - Water Resources Section. (2024). *Standard Operating Procedure - For the Collection of High-Water Marks from a Flood Event*.

Appendix A: Hurricane Helene Flood Photographs



Figure A.1: Flood Damage on bridge in Asheville, NC



Figure A.2: Flood Damage on billboard Asheville, NC

HURRICANE HELENE – POST FLOOD REPORT



Figure A.3: Flood Damage in Swannanoa River near Swannanoa, NC



Figure A.4: Pigeon River floodwaters discharging from Watersville Dam

HURRICANE HELENE – POST FLOOD REPORT



Figure A.5: Damage done to US HWY 421 near Mountain City, TN



Figure A.6: Damage done to Rhodes Road in Edneyville, NC

HURRICANE HELENE – POST FLOOD REPORT



Figure A.7: Damage done to Rhodes Road in Edneyville, NC



Figure A.8: Flood damage at Peppertree Resort in Maggie Valley, NC

HURRICANE HELENE – POST FLOOD REPORT



Figure A.9: Flood damage in recreational park



Figure A.10: Lake Craig Dam Failure on Swannanoa River near Asheville, NC

Appendix B: High-Water Mark Data Collection Table - Example

Table B.1: HWM Field Data Collection Log - Example

Field Map Name	Stream	Event	HWM Type	HWM Quality	Measured Height (ft)	Photos (Y/N)	# of Photos	Survey Date	Description
PR-39-L	Pigeon River	Sep-24	Mud line	Very Poor	0	Y	2	11/Oct/2024	Mud line in left bank of under railroad bridge, not good line, not clear
PR-40-L	Pigeon River	Sep-24	Debris line	Good	0	Y	3	11/Oct/2024	Good debris line in back of house, DS of railroad bridge, lines up with owners photograph he showed, photograph has not been submitted to us yet
PR-41-L	Pigeon River	Sep-24	Seed line	Fair	0	Y	4	11/Oct/2024	Seed line on 507 River Street house and back garage
PR-42-R	Pigeon River	Sep-24	Mud line	Excellent	0	Y	5	1/Nov/2024	Excellent mud line on glass on door on US side of brick building on Champion Dr and 25 ft DS of Champion Dr
PR-43-R	Pigeon River	Sep-24	Debris line	Good	0	Y	4	1/Nov/2024	Good debris line on chain-link fence landward of entrance gate and 30 ft US of Champion Drive and 500 ft from river
PR-44-R	Pigeon River	Sep-24	Seed line	Excellent	0	Y	10	1/Nov/2024	Seed line on wood patio on landward side of home on Champion Dr and 70 ft landward of Champion Drive and 60 ft right of bank of Beaver Dam Creek
PR-45-L	Pigeon River	Sep-24	Mud line	Excellent	0	Y	5	1/Nov/2024	Excellent mud line on riverward side of frame home and 225 ft from river; home on Willow Dr. and 350 ft DS of lower lane of Blackwell Dr
PR-46-L	Pigeon River	Sep-24	Debris line	Fair	0	Y	7	1/Nov/2024	Fair debris line on slope 225 ft landward of intersection of Balsam Drive with Blackwell Dr. and 200 ft US of US lane of Blackwell Dr. Bridge
PR-47-R	Pigeon River	Sep-24	Mud line	Excellent	0	Y	5	1/Nov/2024	Back of Pigeon River Lodge No 386 AF+AM and ~ 9 ft above ground and 145 ft landward of Champion Dr

HURRICANE HELENE – POST FLOOD REPORT

Table B.2: HWM Post-Processing Data - Example

Stream	FIELD MAP NAME	SURVEY DATE	In Field Maps? (Y/N)	SURVEYED?	SURVEY POINT #	NORTHING	EASTING	SURVEYED HEIGHT	POINT NAME (SURVEY)	NEW NAME	POINT # ON STREAM
Pigeon River	PR-39-L	11/Oct/2024	Y	Y	1085	604817.344	2804771.269	1067.722	PR-39-L	PR-34-L	34
Pigeon River	PR-40-L	11/Oct/2024	Y	Y	1087	605041.955	2804852.896	1064.112	PR-40-L	PR-33-L	33
Pigeon River	PR-41-L	11/Oct/2024	Y	Y	1091	606745.275	2803673.239	1057.722	PR-41-L	PR-30-L	30
Pigeon River	PR-42-R	1/Nov/2024	Y	Y	5174	673076.45	856603.398	2589.891	PR-42R	PR-123-R	123
Pigeon River	PR-43-R	1/Nov/2024	Y	Y	5175	672795.111	856658.389	2591.631	PR-43R	PR-124-R	124
Pigeon River	PR-44-R	1/Nov/2024	Y	Y	5176	674356.374	856179.299	2589.533	PR-44R	PR-121-R	121
Pigeon River	PR-45-L	1/Nov/2024	Y	Y	5179	674090.342	855801.557	2588.557	PR-45L	PR-120-L	120
Pigeon River	PR-46-L	1/Nov/2024	Y	Y	5180	673595.357	855814.357	2588.925	PR-46L	PR-122-L	122
Pigeon River	PR-47-R	1/Nov/2024	Y	Y	5182	674854.124	855814.413	2586.947	PR-47R	PR-119-R	119

HURRICANE HELENE – POST FLOOD REPORT

This page intentionally left blank